

A PROJECT REPORT ON

(Report Now: A Mobile Solution for Reporting Road and
Electricity Issues)

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A Project Report Submitted In Partial Fulfillment Of The
Requirements For The Award Of

(Mobil Application) in (data analysis)



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A PROJECT REPORT ON

**Report Now: A Mobile Solution for Reporting Road and
Electricity Issues**

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DECLARATION

I, Asma Abdul Salam Al RAWAHI, I hereby certify that the material given here is original and has not, except for properly recognized sources, been copied in whole or in part from any other source. As a result, all references to previously published material (found in books, journals, magazines, the internet, etc.) are listed in the references or bibliography sections of the primary report.

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APPROVAL FORM

The project report entitled Report Now: A Mobile Solution for Reporting Road and Electricity Issues THE MOBILE APPLICATION submitted by 19f19226 Asma Abdul salam Alrawhi Is approved in partial fulfillment of the Requirements for bachelor's degree in data analysis.

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ABSTRACT

This report aims to outline all the specifications for the project which is called (Report Now: A Mobile Solution for Reporting Road and Electricity). As we frequently encounter roadblocks that some drivers are unable to avoid, which puts their vehicles at risk as well as themselves, this application assists in reporting issues that may impede pedestrians, drivers, or passengers, dealing with these issues quickly, and swifter access to the problem's location and specifics through the appropriate organizations in charge of roads and lights.

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CHAPTER 1 INTRODUCTION

1.1 Overview of the Project:

Despite the world's continued growth and prosperity, which have brought about the highest levels of modernity and renewal and made life significantly easier than it was previously, we still require applications that make daily tasks easier for us, and there are still issues that require prompt attention and solutions, such as the following. It has infrastructural issues because there are several issues that require urgent attention and whose gravity cannot be understated or minimized, This is an issue that cannot be ignored or undervalued since we always see a problem in me. We know that lighting has a huge advantage, especially for drivers at night, as it helps them see the road clearly and also helps them notice if there is any danger on the road. The road is like potholes, and these issues pose a serious risk and may result in accidents; as a result, they must be resolved as soon as possible in order to decrease accidents and ensure the safety of the driver, And now the project's function, which is to report any problems, comes into play. When someone encounters a problem with the lighting or the roads, they can report it using our application by entering the specific information pertaining to the issue and attaching a picture that illustrates it. The application will then forward the issue to the appropriate businesses for resolution within 24 hours, and the person who reported it can check to see if the issue has been resolved.

1.2 problem statement:

We must emphasize this component and pay attention to it so that it is outstanding since there are numerous infrastructure difficulties due to the world's ongoing expansion and wealth, and there is always complacency in these concerns. The community benefits from a high-quality infrastructure and a problem-free environment because we constantly strive to be the best in all that we do and to live as a cohesive community. This ensures that the community is as safe as possible and that everyone can go about their daily lives in safety. As a result, we put into action this project that will lessen issues, provide rapid access to the issue and fix it promptly, and for this project to be a positive contribution to society because we frequently encounter broken street lights. Road users are put in risk because of this, particularly at night. Examples include wandering animals or obstacles in the road that a vehicle cannot see or avoid owing to poor lighting, as well as road issues brought by weather conditions like rain and valleys that endanger both drivers and other road users. The goal behind the initiative is for anybody to enter the program and report information on any issue they see that may impede drivers or other road users (the name of the governorate - the name of the region - a picture of the problem and other details) Once the individual reports the issue, the application will send it to the appropriate firm, such as the road or energy provider, who will then send a team to the location of the issue to resolve it. After that, the specialized company will send a message to the application and inform them if the problem has been resolved or not, and the one who reported the problem can enter the application again after 24 hours and check whether this problem has been resolved or not.

1.3 Project Scope:

Residents in general, especially those who discover issues that no one cares about or in residential areas with a lack of services, are the target audience for the software program. By reporting the issue that has put these drivers at risk, the application will specifically aid these drivers. He will make it clear who is responsible for reporting the issue, along with any accompanying site data and image. The program will then send a message including information about the issue to the appropriate businesses, and those businesses will deal with the issue and find a rapid solution. It is anticipated that the project will be completed in 6 working months. It will begin in March 2023 and end in January 2024 after going through its first stages. In addition to the

anticipated overheads, the overall cost of designing and creating the program application is predicted to be roughly ROM1350. Java was used to create the application, which was connected to the Firebase database as (Back-End) The user interface prototype was developed using Figma and then it was implemented through built using Android Studio using XML (Front - End).

1.4 Project Stakeholders:

Owners are accountable for the project's progress and outcomes. Typically, they consist of project team members, project managers, executives, sponsors from the customer, and users. Stakeholders are those who have an impact on the project at any stage of its development and whose contributions have a direct bearing on the results. Direct and indirect stakeholders are the two categories of stakeholders.

Internal Stakeholder:

The project's first line of defense and those who are immediately impacted by it are internal stakeholders. The project manager, who has direct decision-making authority, the programmer, the company specialist, and the business team of the firm are the internal stakeholders in this project. The project manager is in charge of monitoring and verifying the project's needs overall. He also ensures that the project's progress plan is carried out as planned by the needed plan. There is also the programmer, who converts the specifications into a real application that implements the necessary features and transforms the specifications into functions that can be handled in practice. Additionally, there is the specialized company, which receives the communication and dispatches the work team to the location, as well as the work team of the specialized company (Ministry of Transport, Communications and Information Technology or Municipal), which travels to the location of the issue to address it and then informs the organization if the issue has been resolved or not.

External Stakeholders:

The people or organizations outside the project are known as external stakeholders, and they are the ones who have the most long-term impact over the success of the business or the project. The people or inhabitants themselves make up the project's target audience. Users of the application are also considered external stakeholders

since they report issues with the application and provide information about the issue's location.

Table 1.1 Stakeholder

No	Stakeholder	External/ Internal	communication plan
1	programmer	Internal	Contact directly
2	legislators of the laws and arrangements	External	arrangement a meeting
3	project manager	Internal	Contact via video call
4	Adventurers / Users	External	arrangement a meeting/survey

1.5 Interfaces:

As a user interacts with a program, the user interface is crucial in assisting him.

Android phones are compatible with the application's UI.

The user interface directs the user as they carry out various tasks. A good user interface is one that the user uses quickly and easily. User interfaces make it easier for users to engage with applications.

1.6 Aims and Objectives:

The project's ultimate purpose is to address road and electrical problems, making drivers or road users safer, protecting drivers, offering good, high-quality roads that are free of issues that might lead to accidents for drivers, and correcting problems quickly. the issue Road and electrical concerns were the basis for the creation of an application that was created to be simple to use and a mobile phone application. Goals were established, including:

1- urging motorists to report any issue that prevents them from moving forward will increase road safety.

2- Increasing visibility and safety for drivers through awareness of poor lighting conditions and prompt resolution in order to ensure the security of other road users and to be able to identify any potential obstructions.

3- improving the speed of street problem solving. The companion webpage in the program allows the competent authorities to access the issue site promptly.

4- Accident reduction: Through the changes implemented by the responsible authorities and their assistance in resolving issues, this results in a decrease in the number of accidents.

1.7 Contribution:

The usage of the program will decrease the frequency of street and road issues, make drivers and other road users safer, and improve the condition of the roads. Moreover, the application will benefit road users, specialized businesses, and society at large. It will also promote the community and be a positive project for the community.

CHAPTER 2 INTRODUCTION

2.1 Literature review:

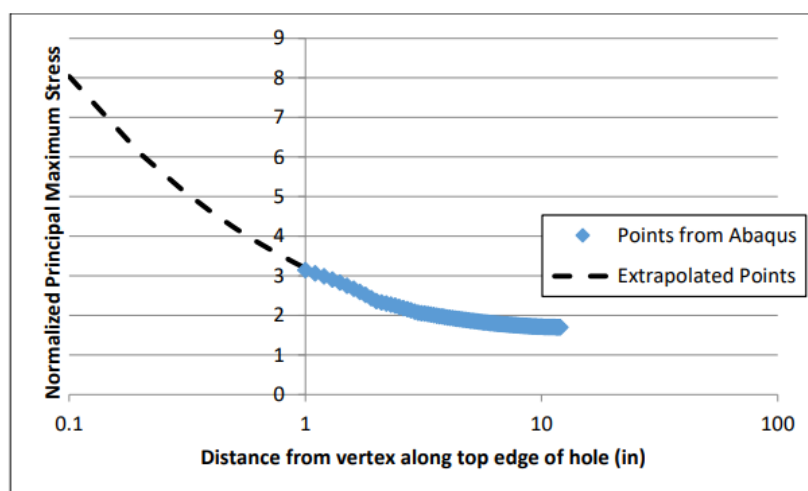
We always find homes in areas that were once semi-deserted due to a lack of infrastructure in some areas and the neglect of the roads, especially in areas where there is a lack of population and where many people do not go, and because remote places are becoming less populated as a result of the increase in population, we must have proper and safe infrastructure, is problem-free, and is safe for users in all places. There are no issues with the roads or the lights.

2.1.1 Shivam khare¹, tanuj verma²(2018) Road deformation suggests flaws as proof of a road or asphalt issue that affects the road's utility, underlying condition, or attractiveness. The term "street deformation" refers to any area of a street, road, or building site that does not meet the requirements for road safety. And having safe streets is one of the most essential conditions for any nation's economic development. Some developed nations invest a lot of money in constructing new streets, and many developing nations have a strong desire to improve their capitalist street systems. Due consideration should be given to the upkeep and upgrading of roadways. Although the term "asphalt" is typically exclusively used to describe the surface layer of a road, it can also apply to the wear path, base course, and sub-base layer. It is a firm, dense outer layer that is positioned over the top of the premium underlayment to provide vehicles with a flat, sturdy surface. Rely on increased capability and behavior for planning purposes.

2.1.2 Seid Diltata Dilgba(2018)The procedures and examinations they conducted revealed that the most common disturbances in the investigation area are alligator fracture, despondency, digging, paddling, edge splitting, longitudinal splitting, and transverse splitting. These are the main topics of this paper, which focuses on the factors that cause street damage. And the ongoing traffic delay is one of these elements that distorts the area under study. We come to the conclusion that maintenance or restoration work is the sector's magic bullet as much as sorting and creating an issue as a result of studying the facts. Users of the streets are

disproportionately affected by street problems. The travel industry's financial growth and future prospects are also impacted by these issues, which lengthen travel times, decrease passenger happiness and well-being, slow down traffic, influence vehicle owners through higher maintenance expenses.

2.1.3 Marasteanu, M. (2018). The main causes of potholes are slow response times to this frequent issue at the beginning of its development and slow road problem repairs. Due to the heavy traffic, cracking along the tracks of the wheels happens from top to bottom and is correlated with the hardness of the asphalt mixture. Most frequently in the form of longitudinal fissures in thin asphalt surfaces. Water penetrates into the pavement surface layer after fractures emerge and are not repaired, speeding up the deterioration process through a variety of methods. Although the majority of issues are located on the pavement's surface, the origins of these issues frequently lie in the foundation of the pavement structure, and fixing these issues can be time-consuming and expensive. In these situations, and to avoid this issue, quick fixes merely postpone the development of craters rather than preventing it. Pothole formation can only be entirely prevented by a combination of a sturdy structural paving system and prompt maintenance procedures.



2.1.4 Van Schalkwyk (2016) The addition of roadway lighting, according to Elvik (1996), lowers fatal crashes at night by 65%, overnight Injury crashes increase by 30%, whereas evening property damage only increases by 15%. As seen in Exhibit, (2004) these percentages differ slightly from those used in Elvik and Vaa

The amount of accidents that occur at night can be decreased by lighting the road, but there are still issues that cannot be resolved. Over the past few decades, There have been advancements in lane marking materials and sign boards. Additionally, the frequency of collisions can be a factor.

Exhibit 10. Illustration of LED adaptive lighting pilot on US 101, Olympia, WA



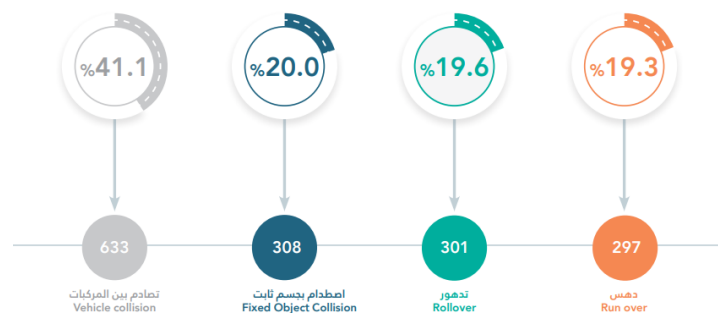
2.1.5 By artificially extending the hours that the light is on so that this is possible activity, street lighting can be utilized to improve security and safety in places and to promote the quality of life. Street lighting offers several benefits for drivers, passengers, or spectators.

Even though only 25% of journeys are made by drivers between 7 p.m. and 8 a.m., same time period accounts for 40% of fatalities and major injuries for the same group of drivers. This is a result of lighting-related impaired visibility. Finding methods that lower the percentage of these risks and safeguard motorists, passengers, and pedestrians on dark roadways is vital. According to a 2003 research by the Ministry of Transport 2, increasing street illumination has a significant positive impact on traffic safety. According to the poll, 63.8% of respondents agreed that "improved lighting of the streets would lead to fewer accidents on the roads," and 73% agreed that "better lighting of the streets would improve safety".

2.1.6 The lack of traditional street lighting, especially at night and at risky intersections and bends, causes dozens of people to die, dozens more to be injured, and we frequently see serious accidents that result in the deaths of many young

people. As a result, the responsible authorities have been working to find solutions to stop this constant loss of life and injury as they always have. As a result of the driver's inability to see well in front of him at night, this issue occurs in locations with inadequate lighting or lighting that is malfunctioning, According to the Sultanate's statistics for the year 2021, the Sultanate recorded 20% of accidents due to collision with a fixed object and 19.6% due to car deterioration. This is especially true because there may be stray animals or something that blocks the road, as well as the holes that exist in the road, which are formed due to various factors, including natural factors like valleys and winds, About 19 accidents as a result of road deficiencies It caused 23 injuries and 19 fatalities, and if the same thing happens again, these numbers will rise. It is urged to address the issue swiftly in order to protect ourselves and others, as well as to lower the number of accidents and fatalities. When someone discovers a problem, their first action is to report it. It needs to be fixed right away to prevent more harm to people and property and to offer a better driving experience.

Figure (4) : Numerical and Proportional Distribution of Traffic Accidents Types 2021



2.3 similar work:

2.3.1 RepotLah: Users of the smartphone app RepotLah may report non-emergency problems including potholes, improper garbage disposal, smoking in restricted areas, and broken traffic lights. You will find the option to "Add new report" right away on the homepage of RepatLha. If you give the app permission, it will automatically tag your current position. Users may then upload a snapshot and report the incident. You can select the specific sort of traffic problem at hand and add a caption to explain the circumstance. After submitting your complaint, you may follow its progress under the "My Activity" page, which gives users access to information about the status of their report. Like Shopee reviews, RepotLah also enables anonymous complaint submission.



2.3.2 FixMyStreet: is a website and app that uses maps created by mySociety that enables individuals the local authorities in the United Kingdom should warn government to issue like potholes and damaged streetlights that require their attention. It is the FixMyStreet UK branch. Although the majority have their own reporting processes for local councils, FixMyStreet addresses the issue that the reporter may not be aware of which authority is in charge of a particular sort of issue in a particular place.[7] FixMyStreet uses the MapIt program to match Postal codes of users and the issue category to the appropriate local authority



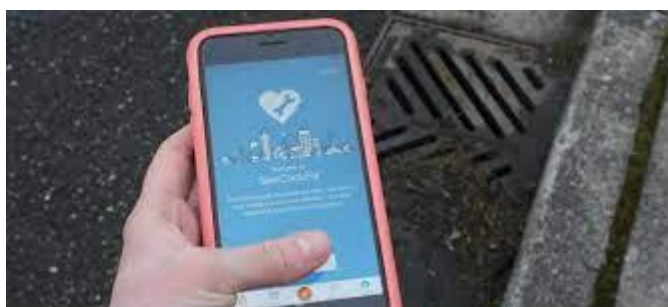
2.3.3 Street Bump: A Smart City initiative called Street Bump seeks to assist locals in enhancing the streets in their communities. While driving, volunteers utilize the Street Bump iPhone app to gather information about the state of the roads. Governments can use the data to plan long-term investments and solve problems in real time. The procedure for gathering data is simple. When you're driving, just open the app and press the "Record a Trip" button. When you end your journey and click the "End Trip

and Upload" option to submit the trip to the government, the information will be posted. When you consider that this is an unintentional bump rather than a true one, you can also cancel the trip by using the "Cancel" button. Smartphones' digital sensors may be able to do their tasks more effectively on their own.

Street Bump, an Android app being tested by the City of Boston, is designed with that in mind. The public should be able to download the software this summer, which uses the smartphone's accelerometer to detect potholes. Street Bump matches information about the magnitude of the bump with a GPS coordinate and sends that to a city database if you hit a pothole while driving while the app is loaded.



2.3.4 SeeClickFix: A 311 request and task management tool called SeeClickFix was created to assist organizations in bridging the gap in communication between the public and local government. whenever a problem is reported it allows specialists to create a ticket and send it to the appropriate department. Users of the program can accept, manage, and route requests for public services, set up workflows, and notify employees of 311 issues. It features an integrated feedback solution that enables administrators to update workers and residents on the progress of work being done. the management of assignments, tracking of customer complaints, issue audits, automated notifications, case management, issue tracking, feedback management, a self-service portal, and routing, and a number of other capabilities are just a few of the many features offered by SeeClickFix.



2.3.5NORTH REPORT: Residents can use the app NORTH REPORT to report problems and track their resolution. It's the quickest and most convenient way to communicate with the City and obtain outcomes. Residents will now find it even simpler to make requests, get acknowledgments, and monitor the status of their tickets thanks to our user experience improvements.



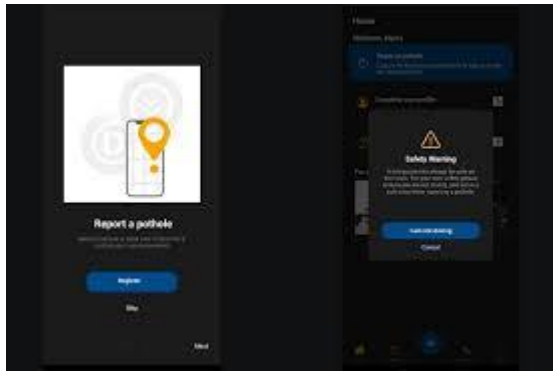
2.3.6Pothole Patrol: The Pothole Patrol was started by Discovery Insure in collaboration with Dial direct and the City of Johannesburg to manage and fix potholes throughout Johannesburg. This innovative data-driven project is in line with Discovery Insurer's primary objective of enhancing and safeguarding people's lives.

You can assist us in creating that favorable change. use the WhatsApp Bot with the Pothole Patrol app to go even further.

The capacity to locate potholes precisely is ultimately what will allow us to repair them. While our unmatched telematics data does allow us to pinpoint potholes fairly precisely through our clients, we can identify a great deal more road faults with the help of the crucial community input. There are now two digital ways available for homeowners and motorists in Johannesburg to report potholes:

1. Get the Pothole Patrol app; it's accessible on Google Play and the iOS app store. You can start reporting potholes as soon as you download the app and complete the registration steps.

2. Text "Hi" to 084-POTHOLE (084 768 4653) on WhatsApp and follow the instructions.



2.3.7 SANRAL Pothole App: The SANRAL Pothole app was created and released in response to the countywide pothole campaign led by Transport Minister Fikile Mbalula. It allows users to report potholes with the touch of a button. Operation Vale Zonke was started by Mbalula to fix the nation's infamous potholes, under strict directives from the minister, (SANRAL) the South African National Roads Agency would take the lead in leading the operation. The maintenance project was given an R11 billion budget that was divided among the provinces. Soon after, the SANRAL Pothole App was released, allowing users to easily log potholes using the app's user interface. SANRAL will take over when you tap to identify the area, give a description, and add photographs. Users will receive updates on how their complaint is going. The software is currently only accessible through the Apple store, which is unfortunate for Android users. There were no results for the app's search on the Google Play store.



2.4 Feasibility Study

It is a study carried out by the project owner on their own or by one of the businesses that specializes in conducting feasibility studies to research a new project or extend an existing project in order to ascertain whether it is feasible to implement the project, its likelihood of success, and the difficulties it will encounter.

the requirements for a successful project in terms of expenditures and resources, as well as its likelihood of success over the short- and long-term. The feasibility study helps determine the project's viability opportunities and potential roadblocks should be identified so that plans can be made to go past them and improve the project's long-term chances of success.

The economic feasibility study and the market research enable one to identify potential hazards in the future and create a backup plan in case they materialize, allowing one to exit with the fewest losses feasible. The costs, expenses, and anticipated revenues of the project are also determined by the feasibility study. Don't invest less time, money, or effort on a feasibility study that informs you of all the project's details before execution and specifies your project's short- and long-term goals. The first step to a project's success is a feasibility study.

2.4.1 Technical Feasibility:

The software was developed utilizing the most cutting-edge technology at the time, including XML for the user interface, Java for the back-end programming to give the application some dynamism, and the Android Studio tool for application creation. Because the software used is a smartphone application, the Firebase database was also used due to its characteristics, such as its ability to send and receive data quickly.

2.4.2 Operational Feasibility:

-There are two login screens; the first checks to see whether the user is new and hasn't registered before, and the second checks to see if the user has.

-The user will have two choices when he signs in regarding the type of report he will submit: will it be about illumination or drilling

-The problem's location and kind are identified when the problem's type is established. A photograph of the problem is also included.

-He receives a message indicating that the communication has been submitted to the appropriate authority to handle the report after completing and validating the necessary information.

-He can also see in the application the resolved and unresolved problems.

2.4.3Economic Feasibility:

Table 2.1 cost:

Cost	Hardware	Software	Human
-	1- Laptop 2- Portable SSD storage 3-Android Phone	Android -1 studio Firebase -2 Database	1- Program 2- repair team 3- nutrition specialist
Total Cost 230+30+85=345 RO	230RO -1 30 RO -2 85RO -3	Free -1 Free -2	Free -1 Free -2 Free -3

Table 2.1 Cost benefit analysis:

The application's advantage to the user is the resolution of issues that might put them at risk from nighttime illumination, as well as guarding, safeguarding lives, supplying a safe route, and lowering the frequency of accidents brought on by these issues.

2.4.4Legal Feasibility:

The project is built on helping the legal authorities by immediately identifying the problem's location and gathering location data to aid in its speedy resolution, respecting the privacy of user information, and offering a service to the neighborhood.

CHAPTER 3 INTRODUCTION

3.1Introduction

In light of the development, prosperity, and progress that the world is witnessing, and the facilities that the world is witnessing, everything has become easy through the Internet and websites that facilitate reports, transactions, and others.... This project aims to provide a safe road for drivers, passengers, and pedestrians, reduce the number of accidents, and create an infrastructure free of potholes and obstacles.

3.1.1Stating the structure

1. methodology: To ensure the project's success and organization, this section will discuss the methodology and framework that were selected as well as the strategy that was chosen for the mobile application's work. It involves a number of procedures, like as investigation, collecting, and analysis, that aid in the completion of the activity. The organizational setting and the project's nature influence the approach chosen. It is crucial that the approach be effective in the project and fit with the nature of the project.
2. Project planning: To ensure the project's success and organization, this section will discuss the methodology and framework that were selected as well as the strategy that was chosen for the mobile application's work. It involves a number of procedures, like as investigation, collecting, and analysis, that aid in the completion of the activity. The organizational setting and the project's nature influence the approach chosen. It is crucial that the approach be effective in the project and fit with the nature of the project.
3. Requirements Analysis: The team is able to comprehend the project, set project objectives, assist in mistake reduction, comprehend the scope and constraints of the project, satisfy stakeholder expectations, assess the project's requirements, record needs, and confirm both functional and non-functional requirements.
4. initial design: develops activities, concepts, and models for the implementation plan, and visualizes the look, path, and visual components of this client interface, including wireframes, templates, and storyboards. This process begins with creating a conceptual plan that outlines how the project will be organized and serves as a starting point as the project progresses.

3.2 SWOT Analysis for Methodology

3.2.1 Prototyping

Let's look at the advantages, disadvantages, opportunities, and threats (SWOT) of the two prototyping methods that were previously addressed.

Strength:

- Early detection of design defects, architectural problems, or usability problems

overall less expensive than creating the completed product-

Using iterative design, versions can be enhanced-

Weakness:

- Prototypes are only partially functioning.

- It requires a lot of resources and time.

Opportunities:

- promotes original design

- evaluating the market's appropriateness and gathering user feedback

- beneficial for luring investment.

Threats:

- irrational hopes based on prototypes

3.2.2waterfall model:

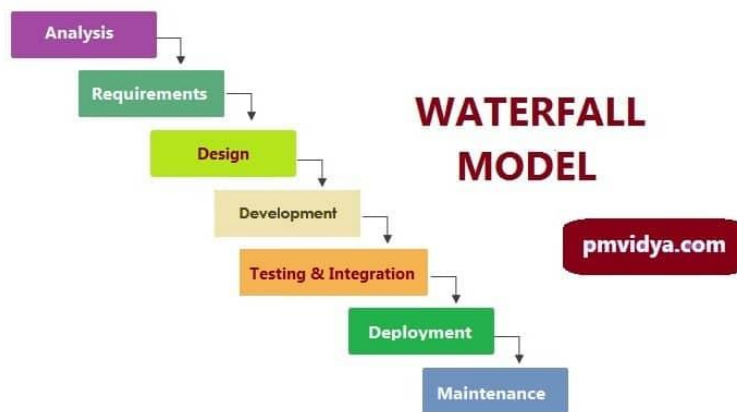


Table 3.1waterfall model:

Strength: - a methodical and sequential approach -A lot of requirements, design, and detail documentation is highly trustworthy.	Weakness: - This model's rigidity makes it challenging to adapt upgrades and adjustments. -It takes a long time to fix faults and errors that are detected too late.
Opportunities: - This paradigm works well for projects with specific criteria and is appropriate for well-defined initiatives. -Documentation and planned phases might be emphasized to assist comply with regulatory standards.	Threats: - The waterfall model finds it difficult to adapt to changes and advancements -Longer development cycles are a result of a waterfall's sequential structure.

3.2.3 spiral model

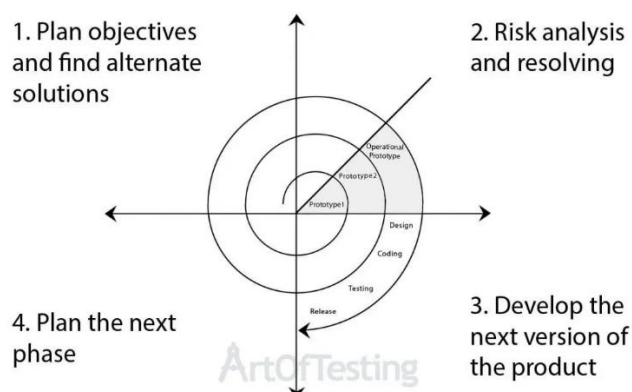


Table 3.2 spiral model:

Strength: -It is affordable. -Recognize dangers early. -use quick prototyping technologies to enable early user interaction. - Making constant improvements in response to user input.	Weakness: -complicated model. -useless for risk assessment need for risk analysis.-
Opportunities: when the model is suitable at the beginning. -For projects for which a risk analysis has been prioritized. -For projects with moderate to high risk. -For tasks whose consumers are unclear about their requirements. -Assistance with the creation of additional related software projects.	Threats: - Limited scalability - Poor communication and collaboration

3.2.4 DSDM Methodology



Table 3.3 DSDM model:

Strength: -Quickly add features to the main product -End users can access it with ease. -Reliable and delivers projects on time	Weakness: - Unsuitable for small enterprises - Costly to implement It can disrupt company culture-
Opportunities: -encourages methodical thinking, which promotes ongoing development. -Organizations can react swiftly to market developments thanks to DSDM. -The DSDM promotes experimentation with new ideas and innovative solutions.	Threats: -Unrealistic expectations regarding the adaptability and speed of DSDM cause incompatibilities and misunderstandings among stakeholders. -Inadequate and insufficient training hinders the effective use of DSDM principles

Table 3.4 methodologies

	iterative	Agile	Linear
Methodology	Spiral	DSDM	Waterfall
Time	long time	short time	short time
Risk analysis	good	good	bad
Quality	high	high	high
Cost	low	medium	low
User involvement	high	high	low
Change control	high	high	medium

3.3 Justifying methodology: iterative

1. Its flexibility and ability to adapt to project changes and developments. Since requirements can change over time, the iterative approach has this flexibility and adaptability, and this methodology offers a framework for incorporating changes at all stages of development.

2- Reducing risks and identifying problems: By breaking the project into smaller iterations and allowing issues to be resolved and addressed earlier in the development process, it is possible to mitigate risks and find problems early on, which helps to prevent problems from arising and to manage risks effectively while facilitating work flow.

3- Improved product quality: By repeatedly testing the product, this approach enables continuous improvement with repetition, which enables problems to be addressed, users' opinions to be taken into account, and the necessary adjustments to obtain a good final product to be discovered and solved in early time.

4- Its capacity to handle shifting priorities: The market's priorities are constantly subject to change. It can adapt to these changes via repetition, and it may be adjusted and structured in accordance with market developments to make it compatible with the market.

5. Feedback and continuous improvement: This involves understanding stakeholders' and users' perspectives, which are then used to inform improvements that are

consistent with users' expectations, increase customer satisfaction, and produce positive outcomes.

6. The iterative process may control the project's scope and expenses through continuous iteration, allowing for the implementation of revisions, resource allocation, and scope avoidance strategies.

3.4 Application methodology

Iterative Methodology Stages:

1. Planning and needs: In this stage, a project plan is established, provided that it is consistent with the project objectives. Difficult needs are also recognized, as well as factors that speed up the project. The first requirements are also established. This phase is useful for gathering information, planning, and creating a project timetable.

2- Design and Analysis: The emphasis here is on the requirements, needs, and execution of the design stage. It is a vital step when the design decides the options to use to achieve the requirements. The greatest design manages to ease financial pressure while delivering the best results. tasks with a time limit for each.

3. Implementation: In order to accomplish the ultimate objective, this iteration is guided by analysis, design, and labor. At this step, needs and plans are applied and coded through analysis and design brought about by the previous stage, and the selected design is built using particular standards and metrics. In order to accomplish the task of designing the system, unit testing is used at every stage of the coding process.

4- Test: It will be examined to see whether it makes more sense after it starts to occur repeatedly. The current iteration of building standards and regulations is also tested at this time (performance testing, pressure testing, safety testing, requirements testing, multi-site testing, etc.), and it is ensured that any flaws have been corrected in order to prevent further errors.

5- Evaluation and review: Following the test, the system's performance is assessed, the factors that need to be changed are properly identified, and the development team, stakeholders, and the authorities in charge of establishing the system analyze the system to determine whether the outcomes satisfy expectations.

spiral model

Testing the spiral model because it is adaptable, compatible with the project's quality and type, and works on risk analysis and significantly lowers the failure rate of large software projects through repeated risk assessment and verification of the product under development at each stage. Additionally, it lessens the likelihood of mistakes and the time required for re-implementation of the spiral model.

Apply the methodology to the project

First, through the first stage, which is planning, the main objectives are achieved in this stage. For instance, defining the project's objectives as well as potential risks, defining the client's requirements and the unit's requirements, and comprehending the system through ongoing communication between the client and and collecting requirements from the user and The competent authorities as well as the project's cost estimate, and Secondly The step of risk analysis comes next. Future hazards are recognized in this phase, and strategies to lessen or minimize them are created. Thirdly, the development stage at this stage as seen by the data gathered. It was gathered from users and organizations responsible for lights and roadways. The software is also put to the test at this point to make sure it's reliable and efficient. A group of users is invited to test the program and monitor its development. The final phase is the review and planning phase, which is a continuation of the previous phase. The first stage is to specify the goals and gather the project's needs from the appropriate authorities and users. An advanced degree of planning occurs in the last phase. The essential tests are run and the materials required to finish the project are chosen for the third iteration. The project is evaluated by users in the fourth iteration.

describing the stages of spiral model:

1. Planning: The essential requirements are gathered at this stage, which also involves estimating the budget, timetable, and resources for iterations as well as the system needs for client and system analyst communication.
2. Risk analysis: The project is looked at in the early stages, prospective risks are found, risk mitigation methods are developed, carried out, and hazards are avoided.
3. Phase of development: This phase involves developing the code and posting the programs on the client's website. The features are developed and tested during this phase.
4. This stage of development entails writing the applications' code and publishing them on the client's website. During this phase, the features are developed and put to the test.

3.5 Project Planning

3.5.1 introduction:

It is a speciality to know how to finish the project in a certain amount of time with a certain number of phases and specified resources. The project's implementation and control phases are outlined in a number of official publications. This stage takes scope, cost, and schedule into account in addition to risk management, resource management, and communication. In order to make sure that plans are thorough and reliable, project planning is employed. In order for the project to succeed, a project plan must be developed. Without a plan, the project would be more susceptible to issues like cost overruns, scope creep, missed deadlines, and other issues. As a result, planning is crucial since it is where all project-related documents such as instructions on how to carry out your project plan, how to manage risks, and any potential

issues—start. All of your project's specifics are included in these documents, which are a component of the project management plan.

3.5.2 work breakdown structure (WBS)

It splits labor into smaller tasks in a hierarchical and visual manner. It is a typical manufacturing technique that facilitates work. One of the most crucial project management papers is the Work Breakdown Structure (WBS), a technical instrument. To guarantee the consistency of project plans, it incorporates the fundamental scope, cost, and schedule by itself. The project team must implement all project work phases in the work organization structure chart, making it a significant planning department that benefits project managers, as well as arrange and define the project's overall scope.

FILE

TASK

TEAM

SHARE

PROJECT

VIEW

FORMAT

REPORT

WINDOW



Undo



Redo



Paste



Cut



Copy

Verdana

B

I

U

abc







11









Outdent

 Free Read Only. Start your free one month trial of Project Plan 365 or Sign In to activate an existing subscription



Calendar



Gantt Chart



Resource Sheet



Risk View



Task Sheet



Enterprise Team Planner



Tracking Gantt



Resource Sheet



Risk View



Task Sheet



Enterprise Team Planner



Tracking Gantt



Network Diagram

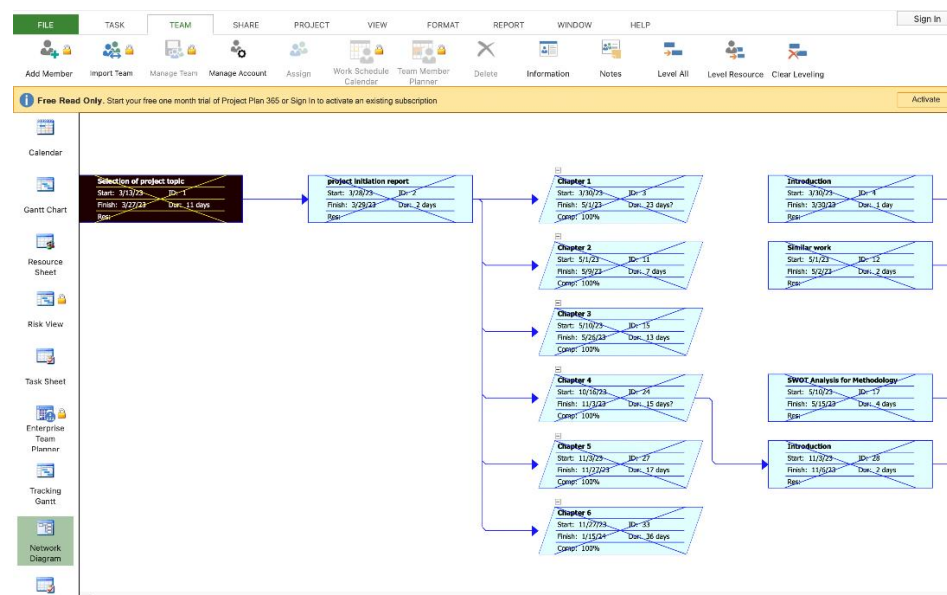
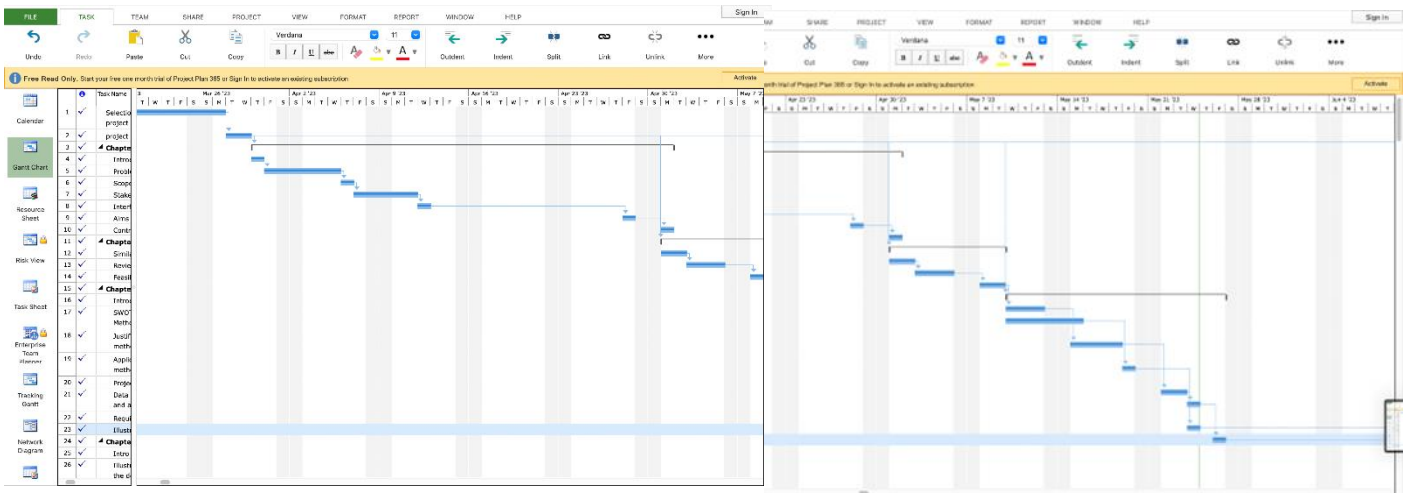


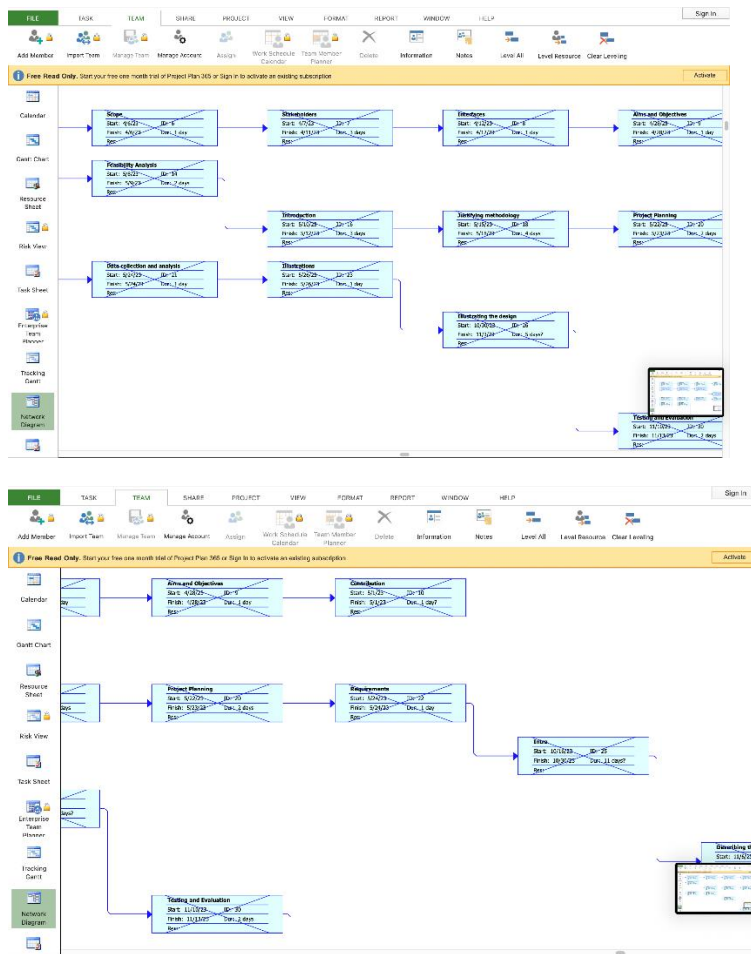
Resource Sheet

		Task Name	Resource Names	Duration	Start	Finish	Work
Calendar	1	✓	Selection of project topic	11 days	Mon Mar 13	Mon Mar 27	6 hrs
	2	✓	project initiation rep	2 days	Tue Mar 28	Wed Mar 29	5 hrs
	3	✓	Chapter 1	23 days?	Thu Mar 30	Mon May 1	46 hrs
Gantt Chart	4	✓	Introduction	1 day	Thu Mar 30	Thu Mar 30	10 hrs
	5	✓	Problem Statemen	4 days	Fri Mar 31	Wed Apr 5	9 hrs
	6	✓	Scope	1 day	Thu Apr 6	Thu Apr 6	7 hrs
Resource Sheet	7	✓	Stakeholders	3 days	Fri Apr 7	Tue Apr 11	4 hrs
	8	✓	Interfaces	1 day	Wed Apr 12	Wed Apr 12	5 hrs
	9	✓	Aims and Objectiv	1 day	Fri Apr 28	Fri Apr 28	6 hrs
Risk View	10	✓	Contribution	1 day?	Mon May 1	Mon May 1	5 hrs
	11	✓	Chapter 2	7 days	Mon May 1	Tue May 9	21 hrs
	12	✓	Similar work	2 days	Mon May 1	Tue May 2	8 hrs
Task Sheet	13	✓	Reviewing literatur	3 days	Wed May 3	Fri May 5	7 hrs
	14	✓	Feasibility Analysis	2 days	Mon May 8	Tue May 9	6 hrs
	15	✓	Chapter 3	13 days	Wed May 10	Fri May 26	60 hrs
Enterprise Team Planner	16	✓	Introduction	3 days	Wed May 10	Fri May 12	10 hrs
	17	✓	SWOT Analysis for Methodology	4 days	Wed May 10	Mon May 15	12 hrs
	18	✓	Justifying methodology	4 days	Mon May 15	Thu May 18	8 hrs
Tracking Gantt	19	✓	Application methodology	1 day	Fri May 19	Fri May 19	9 hrs
	20	✓	Project Planning	2 days	Mon May 22	Tue May 23	6 hrs
	21	✓	Data collection and analysis	1 day	Wed May 24	Wed May 24	7 hrs
Resource Sheet	24	✓	Chapter 4	15 days?	Mon Oct 16	Fri Nov 3	
	25	✓	Intro	11 days?	Mon Oct 16	Mon Oct 30	
	26	✓	Illustrating the design	5 days?	Mon Oct 30	Fri Nov 3	
Risk View	27	✓	Chapter 5	17 days	Fri Nov 3	Mon Nov 27	
	28	✓	Introduction	2 days	Fri Nov 3	Mon Nov 6	
	29	✓	Describing the system	5 days	Mon Nov 6	Fri Nov 10	
Task Sheet	30	✓	Testing and Evaluation	2 days	Fri Nov 10	Mon Nov 13	
	31	✓	Deployment	6 days	Mon Nov 13	Mon Nov 20	
	32	✓	Critical evaluation	6 days	Mon Nov 20	Mon Nov 27	
Enterprise Team Planner	33	✓	Chapter 6	36 days	Mon Nov 27	Mon Jan 15	
	34	✓	Concluding	6 days	Mon Nov 27	Mon Dec 4	
	35	✓	Recommendaton	11 days	Mon Dec 4	Mon Dec 18	
Tracking Gantt	36	✓	Limitation	10 days	Tue Dec 19	Mon Jan 1	
	37	✓	Impact of the Project	10 days	Tue Jan 2	Mon Jan 15	
	Network Diagram						
Resource Sheet							

3.5.3Gantt Charts and Timelines.

You can immediately see what needs to happen and when with a Gantt chart. Along the horizontal axis, time is tracked. The project's tasks are listed together with their start and completion dates and the resources needed on the vertical axis. (Bill Holtsnider2012)





3.5.4 communication plan

A communication strategy is a comprehensive method to planning, outlining, and carrying out marketing, outreach, and communication techniques throughout the departments and teams of a business. The communication plan outlines the information that must be shared, who will get it, when it will be delivered, where it will be distributed, and how it will be monitored.

Table 3. 5 communication plans.

Communication tasks	Communication method	frequency	Objective
Teacher	Teams, Meeting, Email.	weekly	Discuss any requests and queries.
Admin	Email, Teams	Every Day	Supervising the work
student	WhatsApp, Teams	Any time	Discuss application requirements
System analyst	Email	Every Day	Compile project specifications and support stakeholder interaction

3.5.5 Risk management plan:

We approach risk response planning in this manner. It is a step in the process of risk management. The goal of the risk management plan is to recognize, assess, and reduce risks. It also keeps track of risk control and examines costs, returns, and financial consequences. The danger that can be categorized by severity and likelihood the most thoroughly. This is done to identify the effects and establish a plan to lessen them.

Table 3.6 Risk management plan.

	Risk Information	Impact	Mitigation Strategy
1	incorrect or insufficient reporting of traffic issues	hige	In order to assure accurate reports and to incorporate guidance and verification procedures that reduce mistakes, the data verification mechanism's task
2	Lack of user education and knowledge about the application	Medium	Campaign to publicize the initiative, highlight its significance and advantages, and disseminate it in numerous locations.
3	App crashes or technical issues that negatively affect user experience	Medium	Performing quality control tests, seeing mistakes and flaws, resolving them, and developing systems for reporting errors are all ways to promptly discover and remedy problems.
4	Inaccurate reporting of alleged road issues' locations	Medium	To ensure precise position data, use GPS technology and location services. Users should be given visible clues while recording location data. If required, let users change the location manually.

3.5.6 Acceptance Plan

A customer's acceptance of a project's deliverables means that they do so in accordance with the customer's requirements. These requirements are related to the standards of the deliverables, their price, and the deadlines for their production.

Table 3.7 Acceptance Plan.

Task	Acceptance criteria
Design	-Stakeholder approval of the design -Finishing the design documents
Development	-Complete the project's requirements. -Conduct a performance evaluation
Security	-Make a safety check -procedures to guarantee the privacy protection of users

3.5.7 Resource plan

It is a method for finding resources, and by using it, a project may be implemented successfully while also finding the resources it needs. The plan is successful in achieving goals, facilitating the efficient use of resources, and attempting to control risks and resource costs. This strategy will enable better resource management and the achievement of the targeted outcomes.

Table 3.8 Resource plan.

Type of Resource	Resource
Hardware	Computer:programming Mobile phone : to report the problem
Software	Programming tool - Database:send data. -Teams app: To communicate. - Internet
Human	Stakeholder: collect data Project teams: Define the requisite numbers, Consider the users' supervisors.

3.6 Data collection and analysis

The process of gathering data involves finding patterns, trends, and insights. To reach findings and subsequently come to the best judgments, data is gathered, arranged, and evaluated. The goal of data collection is to draw conclusions, make choices, and solve

issues. It involves gathering data from many sources, and after collecting this data, it is processed, including changing the main data, cleaning, and organizing the data. Data collecting is crucial to the process of making decisions that are supported by evidence. It also helps to enhance and streamline existing procedures.

There are primary sources and secondary sources in it.

1 Observations: Internet

2 Interviews: Municipality and the Ministry of Transport and Communications

3 Questionnaires: Internet and Book

3.6.1 Justifying data collection method

One of the most significant scientific studies is the questionnaire. The questionnaire is used to gather the necessary information on the study's subject. It takes the form of a form with questions on it that the respondent must respond to. The questionnaire can be used in a variety of research that seek to uncover facts about existing procedures as well as in polls of the general public and individual perspectives. It is a cost-effective tool with a short shelf life.

It is a quick and simple method to get knowledge, and it is one of the greatest ones.

The goal of the project "Report Now: A Mobile Solution for Reporting Road and Electricity Issues" is to develop a mobile application for smartphones that makes it simpler for people to report problems with the roads and the energy. If you want to ensure the project's success and produce precise and reliable information, it is imperative to have a well-defined data-gathering strategy. This rationale explains why a certain data gathering method was selected and how it meets the objectives of the project:

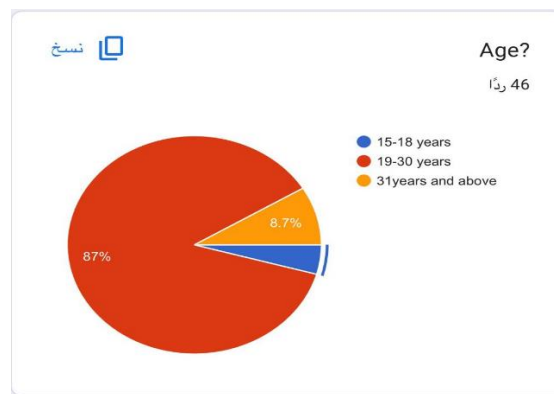
- 1- Real-time Data: Through crowdsourcing, real-time data from users of mobile applications may be directly collected. By employing this technique, anyone may promptly and properly report any electrical or road issues they encounter. The application is made current and valuable via real-time data collection, enabling speedy solutions for found issues.
- 2- User Engagement: By include users in the data collecting process, the crowdsourcing methodology promotes user engagement and provides them the opportunity to improve their communities. Users actively seek out and report electrical and transportation issues, which heightens their sense of responsibility and accountability. This participation also increases the chance of obtaining precise and comprehensive data from those who have dealt with these issues firsthand.
- 3-Wide geographic coverage: Data may be gathered from a variety of areas thanks to crowdsourcing. Because road and power issues might occur anywhere, having a large user base improves the mobile solution's reach and

effectiveness. The method ensures that data is not limited to or biased towards certain locations, leading to a more full knowledge of the larger issues.

3.6.2 Data analysis methods.

Data collection methods is a technique or procedure used for research purposes. There are many outreach and quantitative methods for collecting data. Common methods include questionnaires, interviews, site visits, lab experiments, literature review, documents, etc. These methods can be analyzed and used to support hypotheses. And drawing conclusions about the chosen topic, and choosing the appropriate method for the topic, the type of data required, the resources and the time available.

Questionnaire:



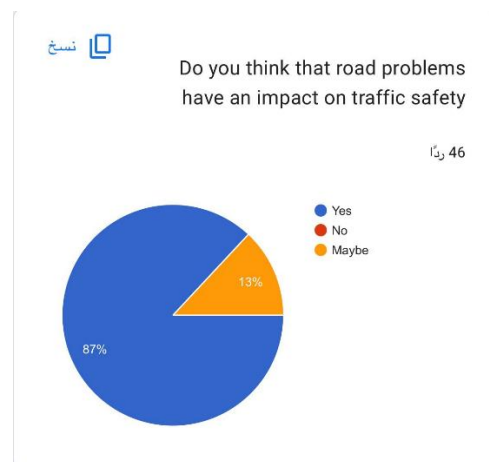
He answered to this survey from various age groups in society, but the majority was from 19 to 30 years, as can be predicted given that this age group is the most engaged in society and represents the young category, which reached 87%.

8.7% of respondents in the group of those over 30 years old and 4.3% of respondents in the category of those between 15 and 18 years old respectively.

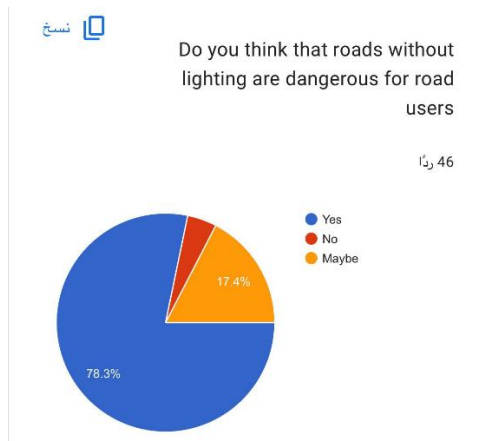


As they come from various and far places, the community members in this inquiry assess the condition of the streets in the area where they reside. This inquiry focuses on the overall condition of the roadways and the degree of safety offered to drivers, passengers, and pedestrians. This allows us to gather data on the streets in a useful way. We are aware that different parts of the nation have streets of varying quality. Some locations are more vulnerable to natural disasters or other circumstances than others, and as a result, some areas see a greater decline in street quality than others. And 46 people responded to this question.

The highest response revealed that 55.5% of respondents thought the quality of the streets was excellent, 26.1% thought it was poor, and 17.4% thought it was very good, highlighting the variation in street quality from one place to another. However, because it has the highest grade, the streets are generally thought to be of good quality.



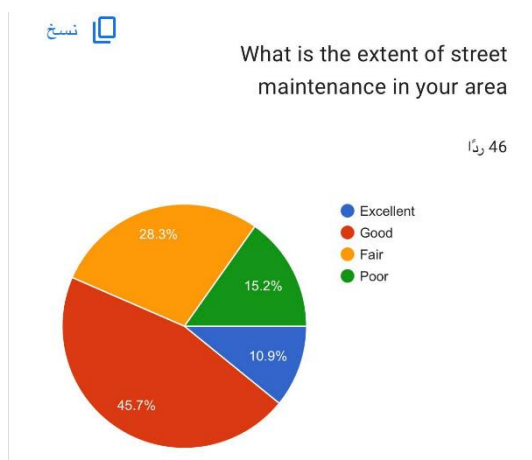
When we asked the community whether they believed there was a connection between traffic safety and road problems, the response was as expected, and the percentage of respondents who agreed with this statement reached 87%. This is the highest number and proof that there is a connection between road problems and safety. This was the anticipated reaction given the traffic. We constantly learn of traffic accidents, injuries, and fatalities as a result of poor road conditions, whether through family members or through social media, and we work to find solutions to these issues to make the mosque a safer place.



In this inquiry, we're trying to determine whether there's a connection between road illumination and user hazard.

There is a risk to drivers, passengers, and pedestrians at night, as seen by the 78.3% of respondents who believed that illumination is correlated with greater risk.

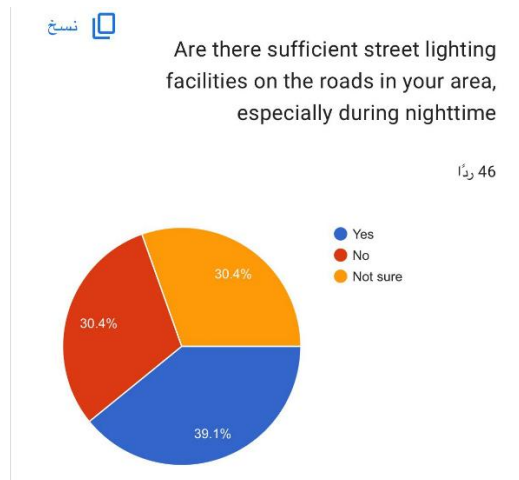
Additionally, we learn about fatal incidents that take place in dark places from others in the neighborhood or on social media. By providing enough illumination on the roadways, the state should work to limit the frequency of accidents involving people and injuries of varied severity.



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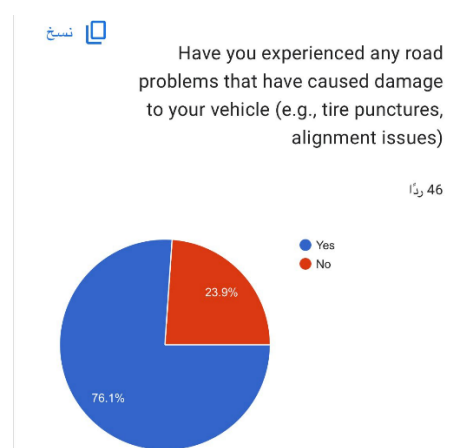
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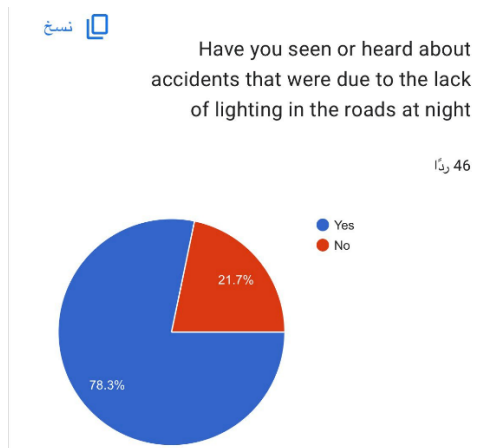


In this inquiry, we're trying to determine whether or not the quantity of illumination on the road is enough and accomplishes its goals.

Positively, Yes received the largest percentage with a rate of 39.1%, while No received a rate of 30.4%. This demonstrates how illumination is distributed differently depending on the region. There are places with adequate illumination and others without it.



In this question, we want to know whether the automobiles had issues with the roads or not. A significant number of 76.1% of the cars had issues with the roads, which is not good and is proof that road issues have an adverse effect on both people and cars.



In this question, we wanted to discover if anyone has experienced or heard of someone experiencing an accident on a dark road. The biggest proportion, at 78.3%, was in favor, which is regrettable and suggests a connection between illumination and accidents.

the interview:

1. How do you think a mobile app can help address road issues?

First of all, reporting road issues like potholes, road damage, or poor illumination is simple thanks to the smartphone application, which offers a simple interface.

Second, the mobile application can pinpoint reported road faults with great precision using the integrated GPS capabilities.

Finally, a mobile application may gather and compile information on reported road issues, building a thorough database of issue spots.

2-How can you verify reported road problems and their significance, and how can you ensure the accuracy and reliability of potholes and reported roads without lighting within the app?

Users can upload photographs of reported road difficulties by asking the website for proof of the issue, such as pictures, which improves the verification process. The severity and significance of potholes or poor illumination are assessed using these visual cues.

Geolocation service provision The application's usage of geolocation services provides a thorough identification of reported road issues. The app's geolocation information allows authorities to confirm reported locations. examining reported road problem data over time offers insight into trends, recurrent problems, and patterns through examining data and trends.

Prioritizing repairs in places where there are many potholes or poorly lit roads can improve the accuracy of complaints reported there.

3-What are the most common problems you face on the roads?

Potholes are depressions or cavities in the road's surface that harm cars, result in collisions, and inconvenience other drivers.

Road maintenance issues Inadequate maintenance can result in worsening road conditions, such as cracked pavements, which has an impact on user experience and traffic safety.

Road users run the danger of being injured or killed on inadequately illuminated roads, particularly at night or in poorly lit locations. When eyesight is impaired by lights, making it harder to see people or obstacles.

4-Are there any safety concerns or incidents related to insufficient street lighting?

Inadequate road lighting impairs drivers' ability to see pedestrians, bicyclists, and other vehicles, which increases the danger of accidents, particularly at night. Injury creates safety problems and increases chances of accidents.

Additionally, poor illumination makes it challenging for both cars and pedestrians to see street signs, obstacles, or other dangers on the road, which can cause navigational issues or accidents.

5-Have you personally encountered any difficulties or incidents while navigating around potholes in your area?

Many individuals have difficulties navigating around potholes due to the fact that potholes may harm cars,

Both cars and pedestrians may be put in danger by potholes. Particularly while traveling at a high speed, sudden contact with deep potholes can result in loss of control or accidents.

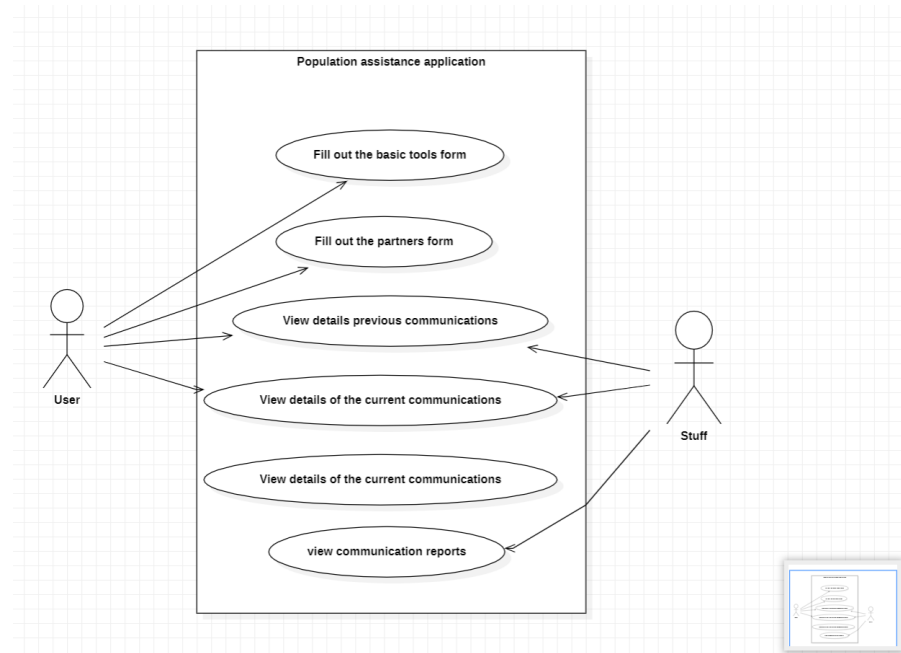
As cars attempt to avoid them, potholes can slow down traffic, causing congestion or abrupt lane changes.

Table: System Requirements

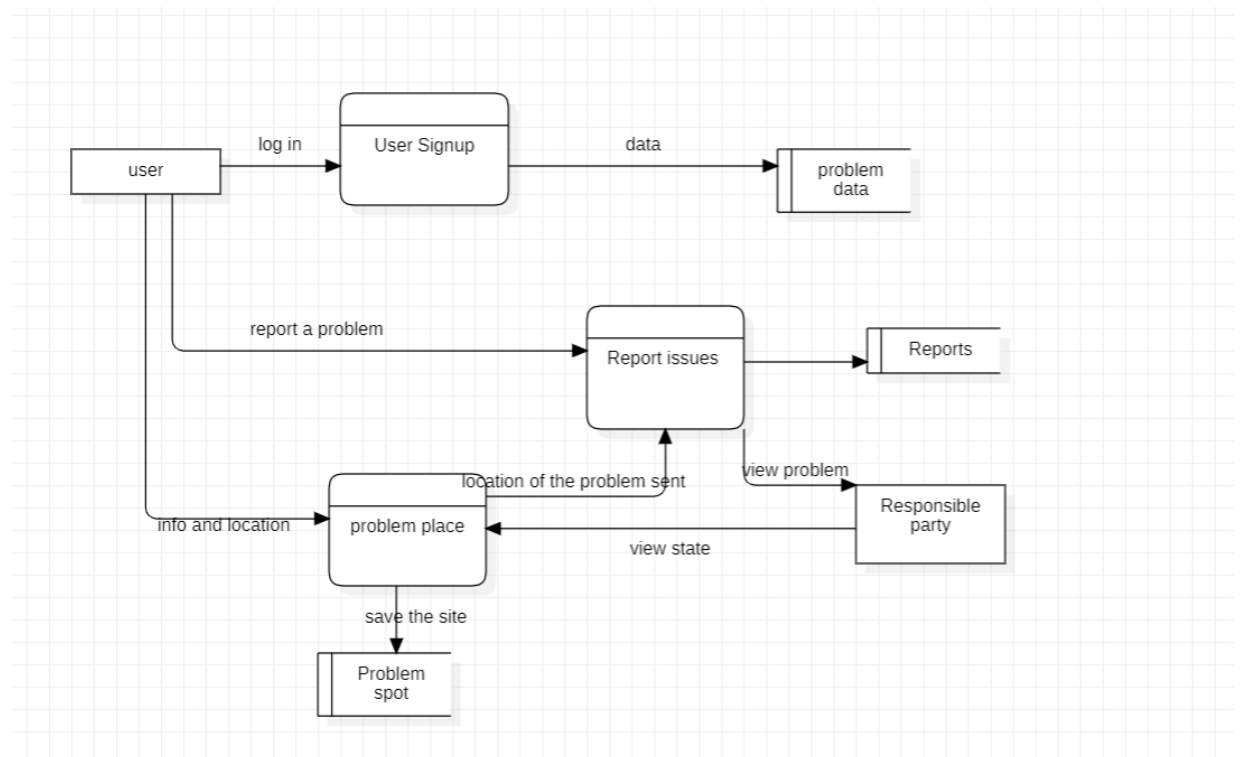
Requirement Category	Requirement Description
External Interface	-A user-friendly interface for the user experience. -Mobile responsiveness: adaptable and device-optimized.
Functional Requirements	-User verification and user signup: the ability to create an account and log in. -Hospital Finder: It should be possible for users to look for hospitals in Muscat using their location.
Performance Requirements	-Scalability: Be able to handle an increasing user base and more data. -Support several concurrent users without compromising performance.
Database Requirements	-Normalization: Make storage of data efficient and well-planned . - Data Migration and Integration: Plan the data transfer.
Security Requirements	-Consistent Security Audits: Identify any systematic issues and address them right away. -ser Authentication: Implement an efficient user authentication process. -Data Privacy: Make use of privacy controls and obtain the required authorization.
Maintainability Requirements	-Code documentation: Completely archive the codebase - Automated Testing: Carry out automated testing techniques like unit testing.
Usability Requirements	-Giving optional content is one of two accessibility considerations. -Quick job completion: Reduce the number of stages and streamline the client stream.

3.7 project requirements

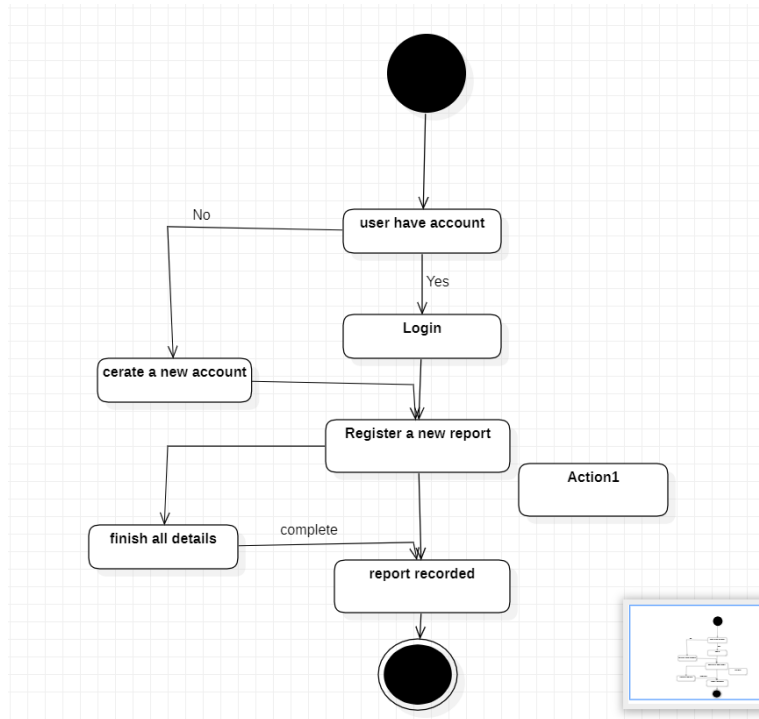
Use case diagram.



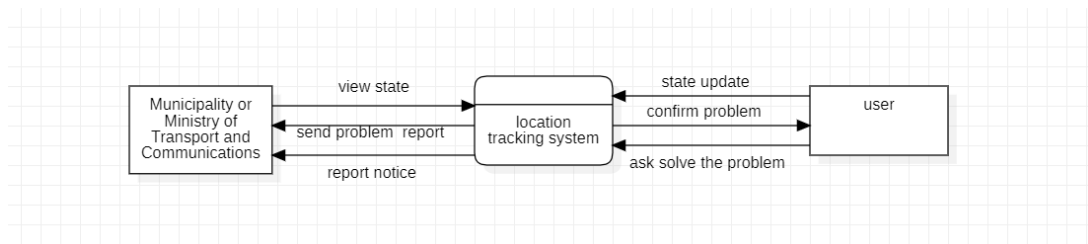
Data flow:



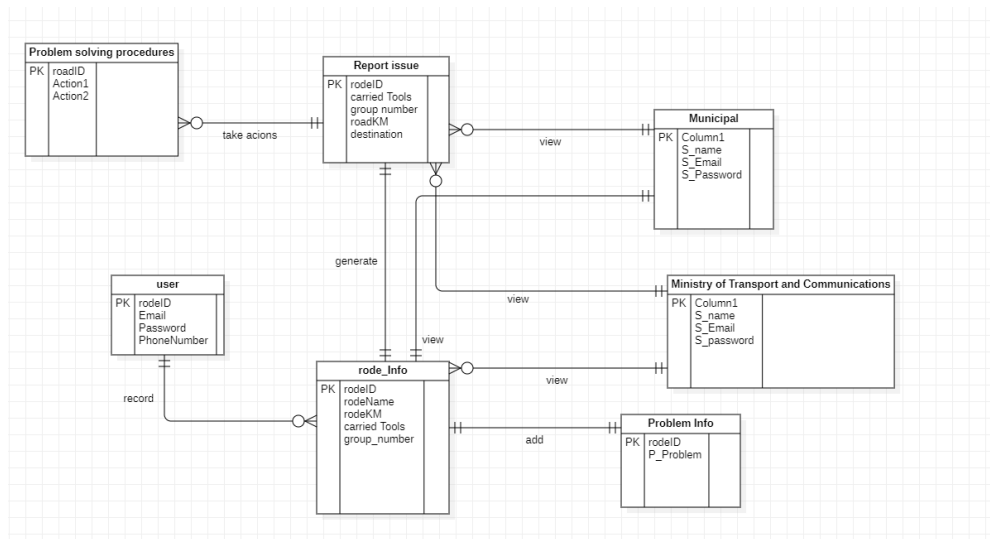
Activity Diagram:



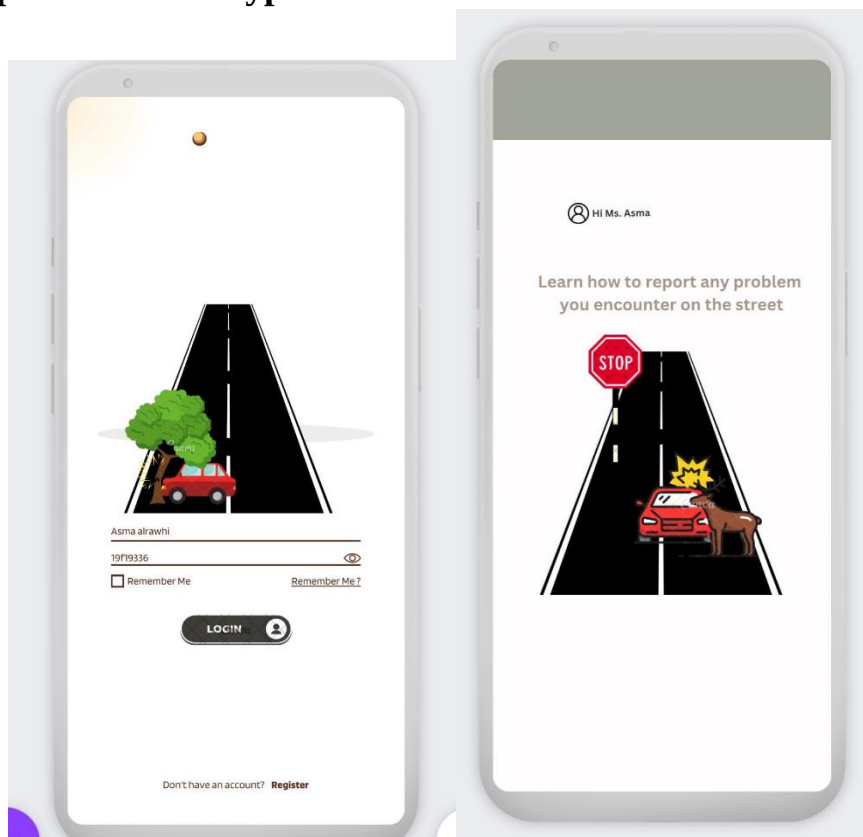
Context Diagram:



Entity Relationship Diagram:



3.8 Application Prototype



CHAPTER 4 INTRODUCTION

4.1 Introduction:

The Adventurer Assist system is thoroughly explained through the use of illustrations in this chapter. using a variety of UML diagrams, including flow chart, sequence, use case, content, and activity diagrams, along with a thorough explanation for each diagram. This chapter also provides screenshots of every component of the system and explains how the data in the system relate to one another as well as how each part functions.

4.2 data dictionary:

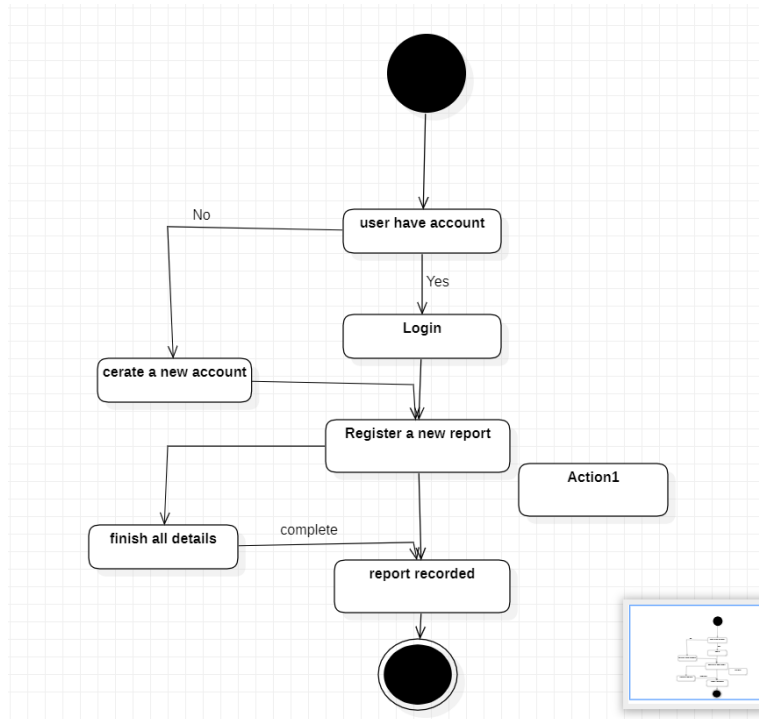
It is an assortment of metadata, object descriptions, and data model elements that programmers and other users refer to. The data dictionary serves as a central repository for metadata and plays a crucial role in data modelling since it facilitates the creation of diagrams. For relationships between objects that are used to list (each object's name, the supplied data values and the supplied relationships) (Chai, 2022). A data dictionary is invaluable for understanding the overall structure, relationships, and flow of the database in addition to helping to identify errors in the data and reduce the time spent finding the data and performing analyses.

4.3 Illustrating the desing:

It is an ornament, a graphical interpretation, a process, a message, or an idea expressed visually. Since illustration can be applied to any technique and convey ideas visually, it is a combination of graphic design and classical illustration.

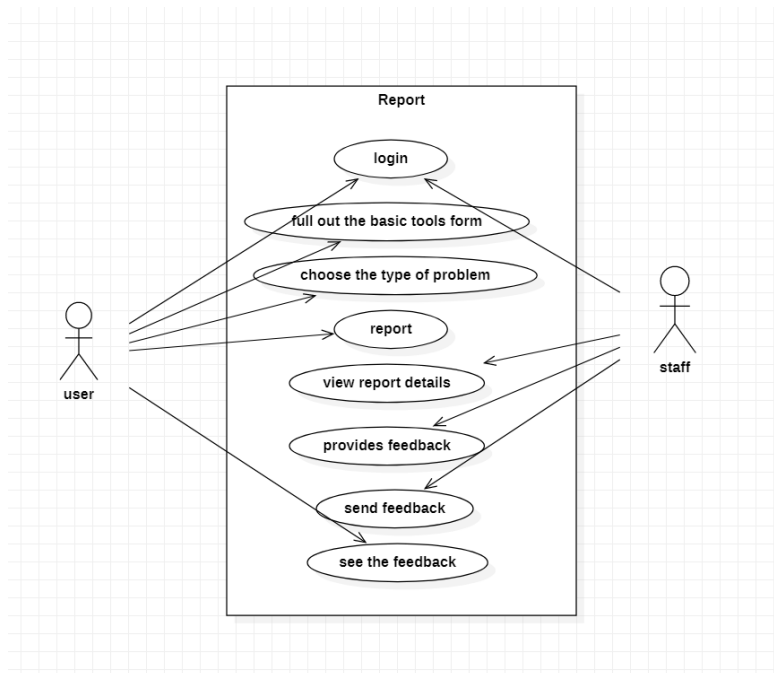
4.3.1 Activitie diagram:

The system development process can make use of the activity diagram, which is process-based rather than object-oriented. It facilitates communication between developers and clients so that both parties can grow and analyse processes, understand workflows, and get a shared understanding of what is happening in the system. It also helps to visualise the system's functions at various levels of detail (Activity Diagram - an Overview | ScienceDirect Topics, n.d.). When a user logs in to the system and fills in the data and details, the activity graph provides a comprehensive explanation of how the system works.



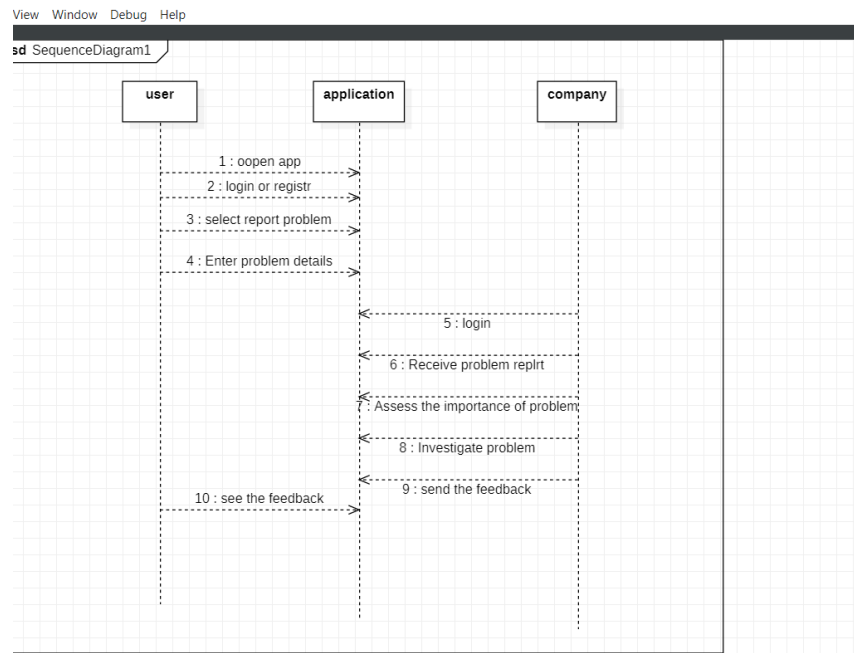
4.3.2 Use case diagram:

The fundamental form of system/software requirements for a new, undeveloped software is called a UML use case diagram. Because the case diagram is straightforward, summarises the relationships between user cases and actors, and incorporates a variety of shapes, it is effective in defining use cases and expected behaviour (Visual Paradigm, 2019). The use case diagram shows how the system is designed, but it also defines the system environment, the user—who is the primary user of the system—and the administrator, who is in charge of managing the application data. In order to report the issue, the user must fill out the required information. Meanwhile, the employee must show the relevant data and reply to the user.



4.3.3 Sequence diagram

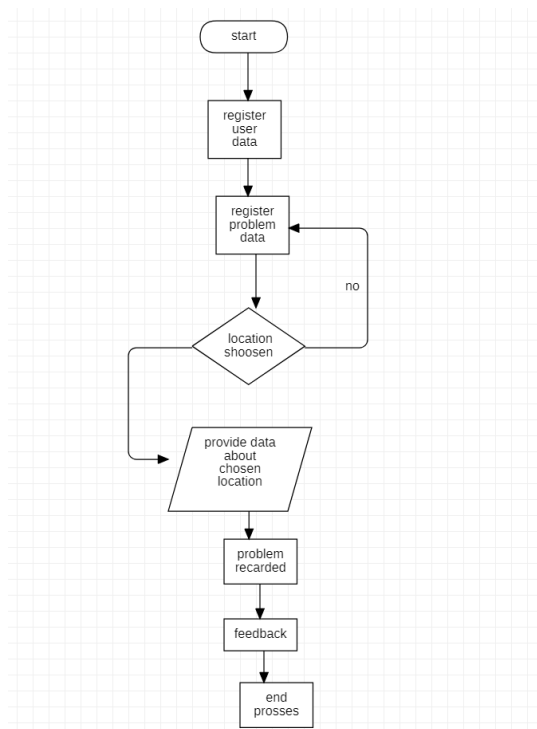
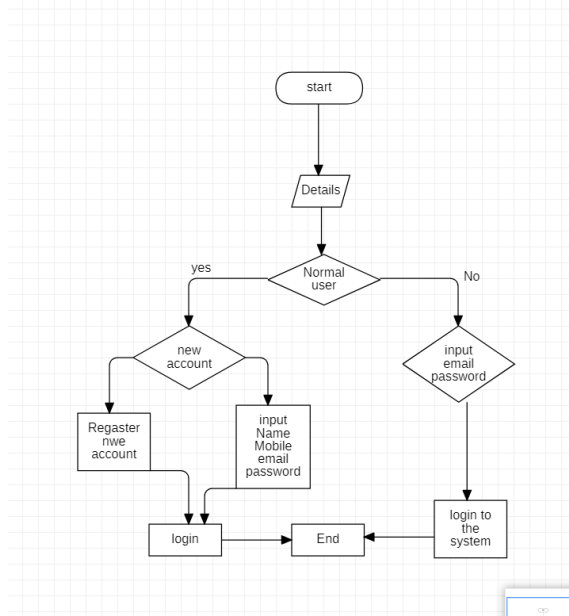
Sequence diagrams display object interactions in a chronological order, making it possible to identify the objects and interactions needed to complete a task by following the flow of events. They strive to provide a step-by-step explanation of what ought to occur to the user. They are helpful in outlining the situation and extracting interactions and requirements from a process. Within an object-driven use case, UML is crucial. Instance roles that depict the states of subsystems, systems, components, or even interacting use cases over time are visualised using sequence diagrams. Two references are included in a high-level sequence diagram for more in-depth interactions. (Sequence Diagram - an Overview | ScienceDirect Topics, n.d.),

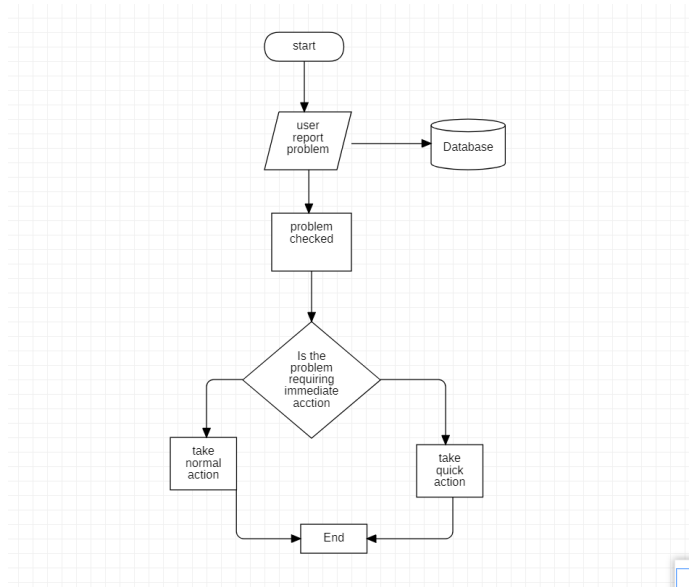


4.3.4 Flow chart:

A flow chart is a diagram that shows the individual steps of a process in a sequential order. It is also a general tool that can be customised for a variety of uses. It is employed to explain various processes, including the project plan, the administrative or service process, and the manufacturing process. A number of chain elements, including steps, supplies, and services that enter or exit the process, are shown in the flowchart. Including (American Society for Quality, 2019: Inputs and Outputs) It describes the steps in the workflow, including logging in, selecting the problem type, filling out the problem data and assessing its significance, and deciding when the problem should be reported to the company so that the employee can view it, assess its severity, and take appropriate action. about the source.

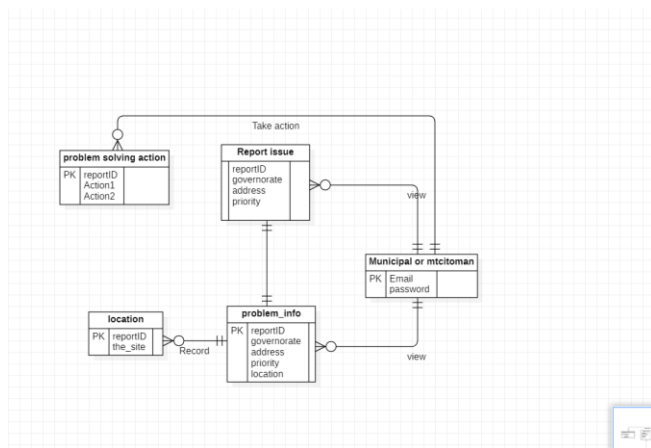
Login page





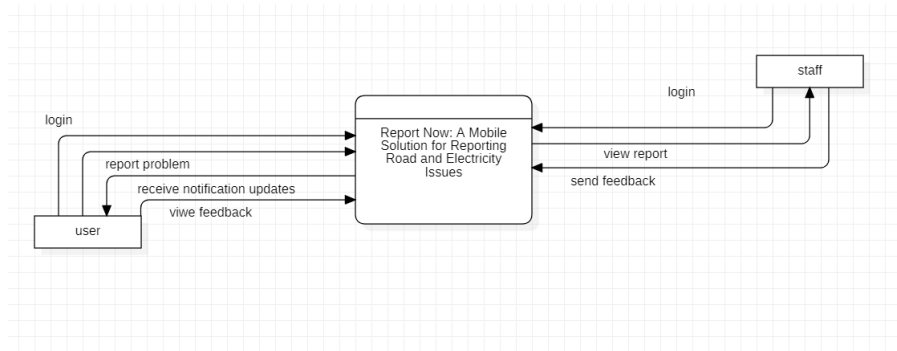
4.3.5 ER diagram

a schematic that illustrates the relationships between various entities, including things, people, and concepts. This diagram will help you create or fix relational databases. In order to connect entities and relationships, symbols like ovals, rhombuses, rectangles, and connecting lines are employed (LucidChart, 2017). Entity relationship diagram .for reference cases to use a NoSQL database



4.3.6 Data flow level 0:

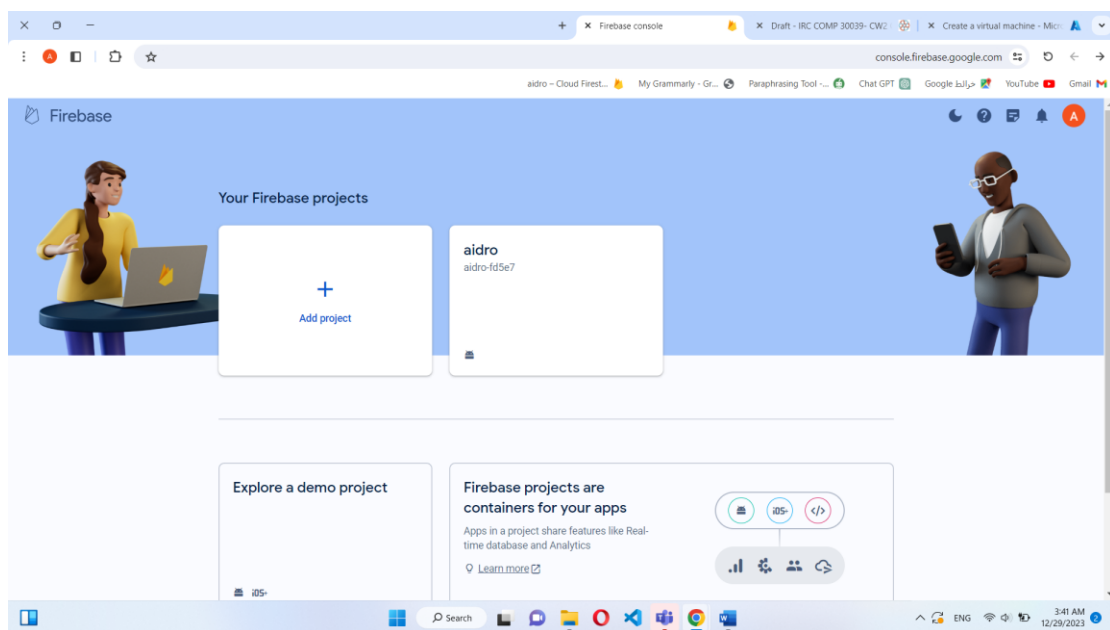
With its system overview, this is the highest level of DFD. Additionally, it makes clear the primary data flows, stores, and processes within the system; however, it omits any information regarding the inner workings of these processes.



4.4 Database:

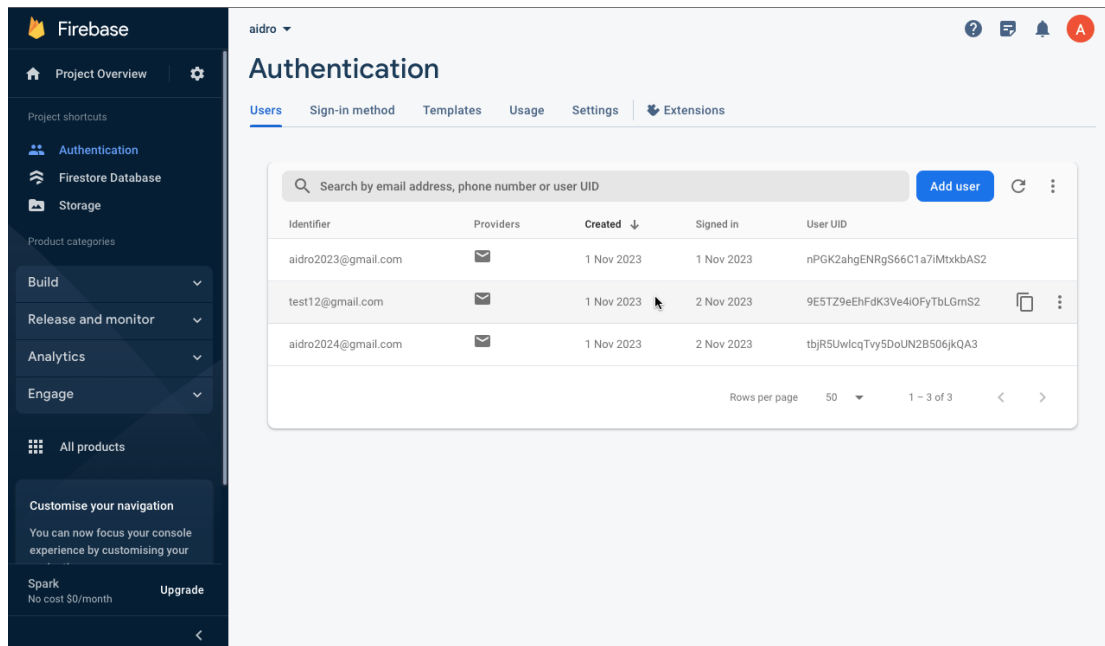
The first step is to create a Firebase registration by signing in to Firebase for the application you made. This registration is secure because it stores the password in a .place where it cannot be hacked and also protects the user's personal data

Firebase Security Rules put a barrier between malicious users and your data. To protect your app's data to the degree that your particular app needs, you can create straightforward or intricate rules(Firebase Security Rules, n.d.).

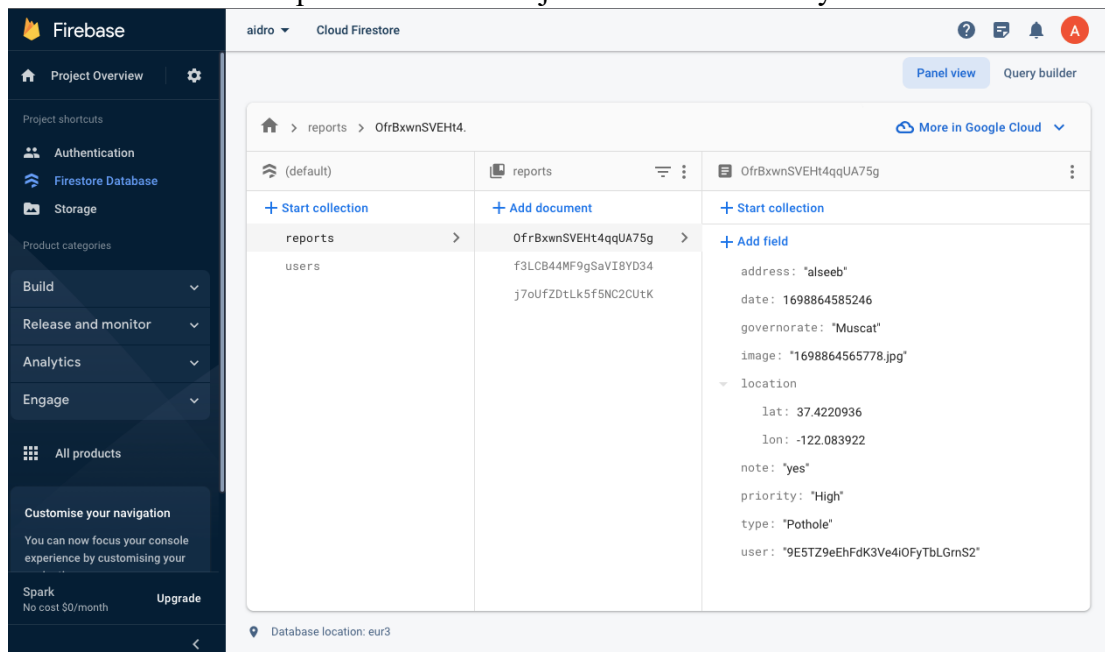


4.6 Describing Security Measures:

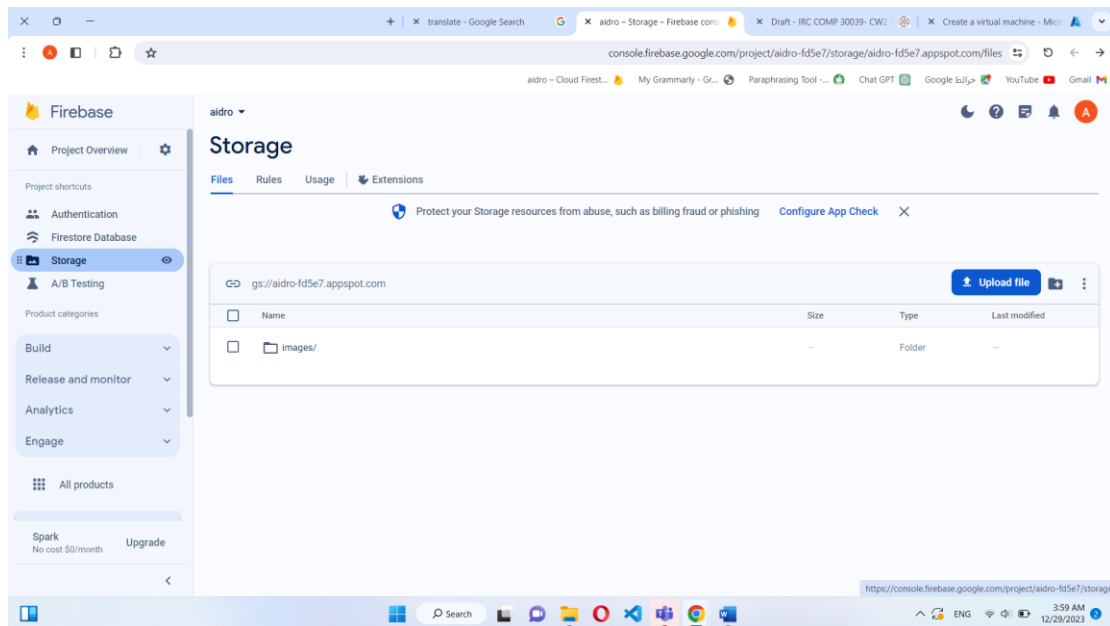
The people who have been notified that the request has been authenticated are displayed in the screen below. This information is displayed for those who have logged into the application, and it can be found in the authorization section of the Firebase page following successful application login.



Firestore is a NoSQL document database that is ideal for application development because of its ease of use, excellent performance, and automatic scalability. Though it is a NoSQL database, the Firestore interface is similar to traditional databases in that it describes relationships between data objects in a different way.



Object storage made to grow with Google is made easy, affordable, and powerful with Cloud Storage for Firebase. Google Security is now added to file uploads and downloads for Firebase apps using the Firebase SDKs for Cloud Storage.



The Metadata Stored: Firebase Database Implementation:

To store user data and login conditionals we use Firebase Firestore database which store data in a JSON format using No-SQL technologies. User password and emails are encrypted and stored using Firebase built-in authentication methods, so we only save user data such as name and mobile in our database linked with user Id that is s email and password in Firebase authentication.'stored along with user

User Information Table:

Field	Type	Description
Name	String	This input for user full name
Mobile	Number	This to store user mobile number
Email	String	This is for the user email.
Mobile	String	This is for the user's phone number

Table 1 – Users Table

Field	Type	Description
Id	String	User ID
Type	String	This is to choose the type of problem
governorate	String	This is to choose the governorate in which the problem exists
Address	String	This is to write the address of the problem
Image	String	This is to take a picture of the problem
Location	Object	This is to determine the location of the problem
Note	Boolean	This is for writing any problem note
User	String	To know the user
Date	Date	This is to know the date the problem was reported

4.6 Describing Security Measures:

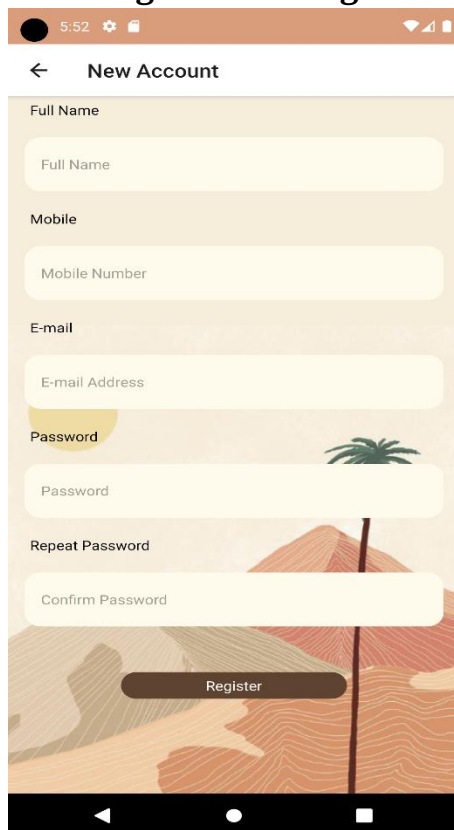
The app supports both iOS and Android and is organised in Android Studio and Visual Studio using the JavaScript language (React Native). All of the app's data is saved in Firebase. The goal of Firebase is to offer a complete method for creating, enhancing, and developing mobile applications and websites. Additionally, it functions to synchronise and store data for every user. Additionally, it enhances the functionality and features of your user experience. Additionally, it has security vulnerability and error fixes that protect user data. Strong authentication services are provided by Firebase. It accepts a variety of login credentials, including passwords and emails. Moreover, Firebase gives developers the ability to designate security guidelines for their databases, control who can read and write data, and specify the prerequisites for reading.

Information technology security has several objectives, such as preventing internal and external threats and safeguarding these resources, tools, and services against misuse, theft, or unapproved use. In order to minimise vulnerabilities and prevent threats, this strategy employs a number of techniques, one of which is the detection

and prevention of security threats. To earn customers' trust, it is important to implement a risk management strategy pertaining to security and privacy.

In order to lower the risk of privacy violations, this application operates with high privacy, gathering only the information that is required and avoiding redundant data. Second, I have implemented controls that prevent unauthorised employees from accessing user data, and I have created a unique email address and password that only authorised users can use to access the application. Third, incorporate privacy principles into the application's design. The application can establish a robust privacy framework, improve user trust, and ensure compliance with privacy regulations by incorporating design principles and subsequently integrating these privacy measures.

5.2.1 Registration Page

A mobile application registration page titled "New Account". The page features a series of input fields for "Full Name", "Mobile Number", "E-mail Address", "Password", and "Repeat Password" (labeled "Confirm Password"). A "Register" button is positioned at the bottom. The background is a stylized illustration of a desert landscape with a palm tree and pyramids. The top status bar shows the time as 5:52 and various icons. The bottom navigation bar is visible at the very bottom.

5:52

← New Account

Full Name

Full Name

Mobile

Mobile Number

E-mail

E-mail Address

Password

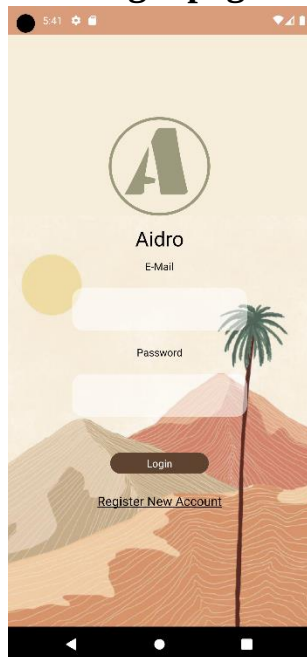
Password

Repeat Password

Confirm Password

Register

5.2.2 Login page



5.2.3 Home Page



5.2.4 Report page

5:52

← Report

Pothole

Choose the governorate

Muscat

Address

Address

Priority

Low High

Note

Note

Picture of the problem Problem Location

Capture Get Location

Send

5.2.5 Login damin

5:41



Aidro

E-Mail

Password

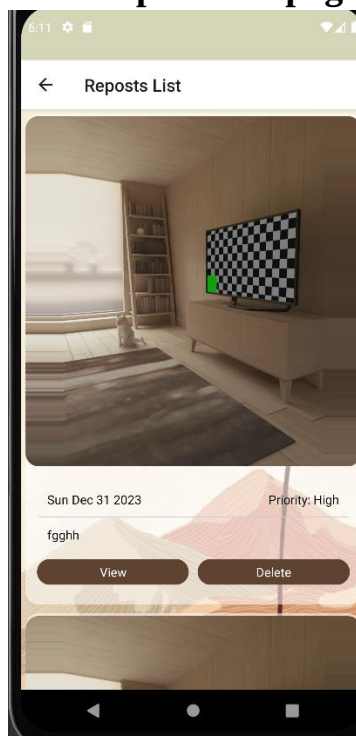
Login

[Register New Account](#)

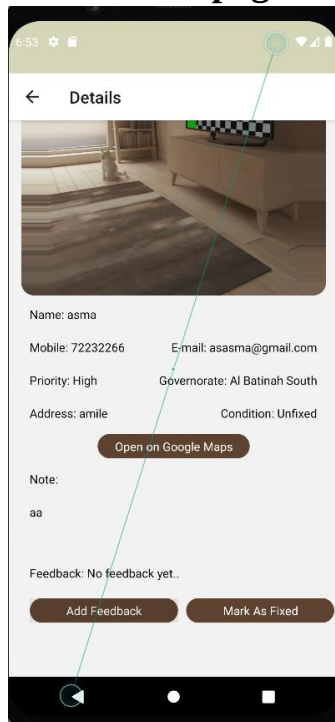
5.2.6 Home admin page



5.2.6 Reports list page



5.2.7 Details page



CHAPTER 5

5.1 Introduction

The development and assessment of each activity and function included in the application are covered in detail in this chapter. This chapter also provides a description of the system and explains how each component works. It also assesses the system's efficacy and tests it through various phases. To guarantee the correctness and quality of the data flow between the system and the database, each system function is examined independently in addition to the system testing procedures.

5.2 System description

5.2.1 Registration Page

In the event that the user is brand-new and has never used the application before, they can register in the system by visiting the registration page and entering their information there. You will be able to use the system functions by accessing the home page if the data is accurate.

5:52

← New Account

Full Name

Full Name

Mobile

Mobile Number

E-mail

E-mail Address

Password

Password

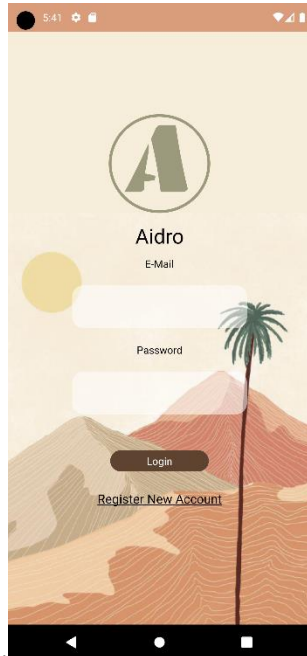
Repeat Password

Confirm Password

Register

5.2.2 Login page

In the event that the user has already logged in, the application's login page has text fields, a login button, and a new user registration button. The user enters his or her email address and password in the text fields to gain access to the application's home page



5.2.3 Home Page

On this page there are the main activities of the system, which are reporting a problem with potholes in the street, reporting a problem with street lighting, feedback, and the logout button

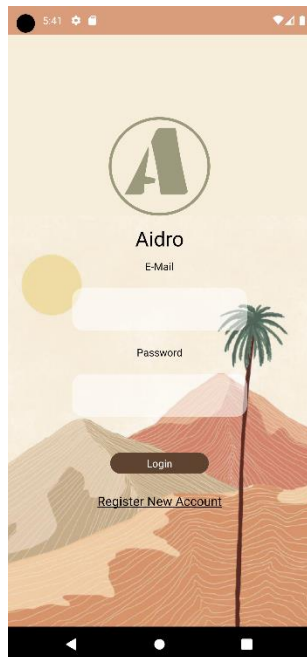


5.2.4 Details page

To report the issue, fill out the necessary information on this page. The governorate, address, location, and problem image are the necessary data.

5.2.5 Login admin.

This is the admin login page. There is a dedicated email and password that only you can enter through.



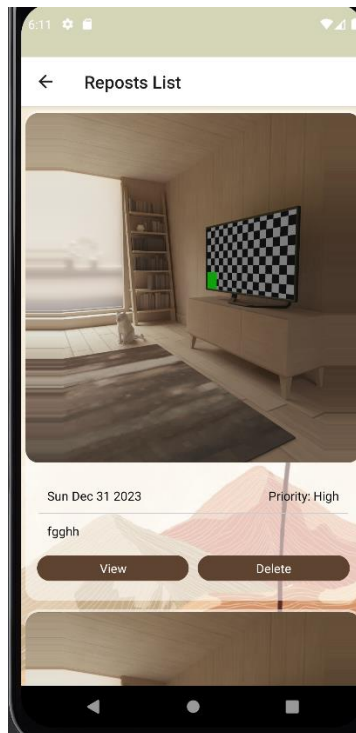
5.2.6 Home admin page

This is the home page in the admin application.



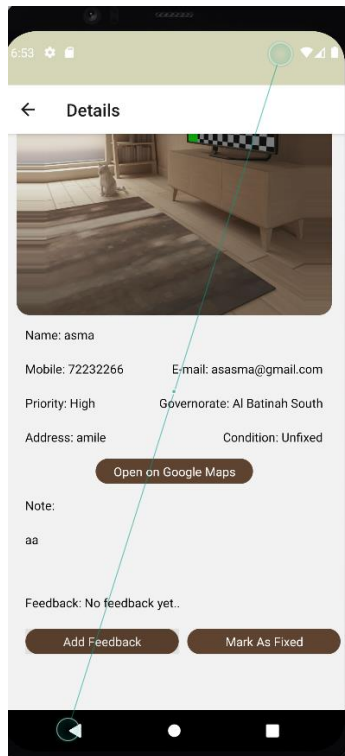
5.2.6 Reports list page

This is a page where all reports can be viewed.



5.2.7 Details page

On this page, you can view the details of the problem, write a feedback, and send it



5.3 coding

Code Implementation

This section has some of the codes that have been used to create this application. Where it had been created by using JavaScript with React Native framework.

```

15
16 const Stack = createNativeStackNavigator();
17
18 function App() {
19   const [initializing, setInitializing] = useState(true);
20   const [user, setUser] = useState();
21
22   // Handle user state changes
23   function onAuthStateChanged(user) {
24     setUser(user);
25     if (initializing) setInitializing(false);
26   }
27
28   useEffect(() => {
29     const subscriber = auth().onAuthStateChanged(onAuthStateChanged);
30     return subscriber; // unsubscribe on unmount
31   }, []);
32
33   if (initializing) return null;
34
35   if (!user) {
36     return (<SafeAreaProvider>
37       <StatusBar
38         backgroundColor="#d5d5b7"
39       />
40       <NavigationContainer>
41         <Stack.Navigator>
42           <Stack.Screen options={{headerShown: false}} name="Login" component={LoginScreen} />
43           <Stack.Screen options={{headerShown: true, title: 'New Account'}} name="Register" component={Register} />
44         </Stack.Navigator>
45       </NavigationContainer>
46     </SafeAreaProvider>);
47   }
48

```

Figure 1: App.js - Here we initiate react native navigation stack so we can switch between meltable screen an activities, we check if the user is not logged in then we only allow to navigate to Login activity screen and the registration screen.

```

49   return (
50     <SafeAreaProvider>
51       <StatusBar
52         backgroundColor="#d5d5b7"
53       />
54       <NavigationContainer>
55         <Stack.Navigator>
56           <Stack.Screen options={{headerShown: false}} name="Home" component={HomeScreen} initialParams={{user:
57             user}} />
58           <Stack.Screen options={{headerShown: true, title: 'Feedback'}} name="Feedback" component={Feedback}
59             initialParams={{user: user}} />
60           <Stack.Screen options={{headerShown: true, title: 'Reposts List'}} name="ReportsList" component=
61             {ReportsList} initialParams={{user: user}} />
62           <Stack.Screen options={{headerShown: true, title: 'Report'}} name="Report" component={ReportScreen}
63             initialParams={{user: user}} />
64           <Stack.Screen options={{headerShown: true, title: 'Details'}} name="Details" component={Details} />
65         </Stack.Navigator>
66       </NavigationContainer>
67     </SafeAreaProvider>;
68   }
69
70   export default App;

```

Figure 2: App.js - In this code snippet we continue to see if the user is logged in then we display the screen for our application.

```

15  const LoginScreen = ({navigation}) => {
16    const [userType, setUserType] = useState('user');
17    const [msg, setErrorMsg] = useState(null);
18    const [email, setEmail] = useState('');
19    const [password, setPassword] = useState('');
20
21    const handleLogin = () => {
22      setErrorMsg(null)
23      if(email !== '' && password !== ''){
24        auth()
25        .signInWithEmailAndPassword(email, password)
26        .then(() => {
27          //console.log('User account created & signed in!');
28        })
29        .catch(error => {
30          setErrorMsg("wrong email or password.")
31        });
32      } else {
33        setErrorMsg("Please enter you email and password.")
34      }
35    };

```

Figure 3: Login.js - here we use Firebase Auth function to log the user to the application if they have an account after we check that they inserted their email and password correctly


```

16  const handleRgeister = () => {
17      setErrorMsg(null)
18      if(password !== rpassword ) {
19          setErrorMsg("Password are not matched")
20      } else if(name !== '' && email !== '' && password !== ''){
21          auth()
22          .createUserWithEmailAndPassword(email, password)
23          .then(() => {
24              firestore()
25              .collection('users')
26              .add({
27                  name: name,
28                  email: email,
29                  mobile: mobile,
30                  userType: 'user'
31              })
32              .then(() => {
33                  //console.log('User added!');
34              });
35          })
36          .catch(error => {
37              if (error.code === 'auth/email-already-in-use') {
38                  setErrorMsg('That email address is already in use!');
39              } else if (error.code === 'auth/invalid-email') {
40                  setErrorMsg('That email address is invalid!');
41              } else {
42                  setErrorMsg("Something went wrong, try again!");
43              }
44          });
45      } else {
46          setErrorMsg("Please enter all required fields")
47      }
48  };
49

```

Figure 4: Register.js - in this code we register the user after insure that they entered their required information correctly, we also check if the account is already existed or the email is invalid, for this we use Firebase build-in Auth function to perform this action.

```

15  const HomeScreen = ({route, navigation}) => {
16    const [loading, setLoading] = useState(true)
17    const [userType, setUserType] = useState('user')
18    const [name, setName] = useState('')
19    const { user } = route.params;
20
21    useEffect(() => {
22      firestore()
23        .collection('users')
24        .doc(user.uid)
25        .get()
26        .then(doc => {
27          setLoading(false)
28          setUserType(doc.data().userType)
29          setName(doc.data().name)
30        })
31        .catch(() => setLoading(false))
32    }, [])
33
34    const handleLogout = () => {
35      auth()
36        .signOut()
37        .then(() => {
38          //
39        });
40    }
41

```

Figure 5: Home.js - in this code we load user information so we display the menu according to their user type if they are admin the we display submitted reports list and if they are a normal users then we let them report issues and get feedbacks for their reports.

```

79     const handleReport = () => {
80         const reoprt = {
81             type: reportType,
82             governorate: governorate,
83             address: address,
84             image: image,
85             location: location,
86             priority: priority,
87             note: note,
88             user: user.uid,
89             date: Date.now()
90         }
91
92         if(address !== ''){
93             firestore()
94                 .collection('reports')
95                 .add(reoprt)
96                 .then(() => {
97                     navigation.navigate("Home")
98                 });
99         } else {
100             setErrMsg("Enter the Address.")
101         }
102     }

```

Figure 6: Report.js - here after getting all the user entered data we save this information to Firebase database and we navigate to the home screen afterward. We also link the issue to user id so we later can know the user who reported this issue.

```

7  const Feedback = ({ route, navigation }) => {
8      const [data, setData] = useState(null)
9      const { user } = route.params;
10
11      useEffect(() => {
12          firestore()
13              .collection('reports')
14              .where('user', '==', user.uid)
15              .orderBy('date', 'desc')
16              .get()
17              .then(querySnapshot => {
18                  let temp_arr = []
19                  querySnapshot.forEach(documentSnapshot => {
20                      temp_arr.push({
21                          id: documentSnapshot.id,
22                          type: documentSnapshot.data().type,
23                          comment: documentSnapshot.data().comment,
24                          image: documentSnapshot.data().image,
25                          date: documentSnapshot.data().date,
26                      })
27                  });
28
29                  setData(temp_arr)
30              })
31              .catch(err => {
32                  //
33              })
34
35      }, [])

```

Figure 7: Feedback.js - here we display all the issues reported by the user and that are linked to their id, we display all the issues to the user and they can see if they got any feedback and read it accordingly..

```

7  const ReportsList = ({ route, navigation }) => {
8      const [data, setData] = useState(null)
9      const { user } = route.params;
10
11      useEffect(() => {
12          firestore()
13              .collection('users')
14              .doc(user.uid)
15              .get()
16              .then(doc => {
17                  firestore()
18                      .collection('reports')
19                      .where('type', '==', doc.data().section)
20                      .orderBy('date', 'desc')
21                      .get()
22                      .then(querySnapshot => {
23                          let temp_arr = []
24                          querySnapshot.forEach(documentSnapshot => {
25                              temp_arr.push({
26                                  id: documentSnapshot.id,
27                                  type: documentSnapshot.data().type,
28                                  address: documentSnapshot.data().address,
29                                  image: documentSnapshot.data().image,
30                                  priority: documentSnapshot.data().priority,
31                                  date: documentSnapshot.data().date
32                              })
33                          });
34
35                          setData(temp_arr)
36                      })
37                      .catch(err => {
38                          //
39                      })
40              })

```

Figure 8: ReportList.js - in admin section in Aidro application we display all reports submitted by users according to the admin user section if they are Pothole managers or they are a Lighting issues administrators then they can display and react by giving feedback to the users.

```

15  const handleComment = () => {
16      if(comment !== ''){
17          firestore()
18              .collection('reports')
19              .doc(item)
20              .update({
21                  comment: comment,
22              })
23              .then(() => {
24                  setShow(false)
25              });
26      } else {
27          setErrorMsg('Please enter the comment text.')
28      }
29  }
30

```

Figure 9: Details.js - when and admin submit a feedback we update the report and we add the feedback to the report so user can followup with the report.

5.4 Testing and Evaluation

5.4.1 importance of testing

One of the most crucial phases of a project involving software products is software testing. Software testing is the process of assessing software to ensure that it is error-free, that it operates and behaves as needed, and that it produces the desired results. The most significant advantages of software testing for software production processes are as follows:

- Prevent errors
- Lower software development costs
- Improve software performance.
- Increase customer satisfaction.

5.4.2 Testing Strategy

are various methods that can be used to test and gauge the quality of a software product. Each strategy also has a distinct objective and set of analytical findings that aid in the necessary development of the software product. Here are a few techniques for evaluating software product quality and making sure it performs as needed:

- Integration test: This test verifies that all software components function together as intended, without glitches or failures.
- Functional test: This test verifies that the functions are present and operating as intended and tests the functional requirements.
- System test: This test looks at the software to make sure it complies with the.

5.4.3 Providing information in tables/figures.

Report Now: A Mobile Solution for Reporting Road and Electricity Issues						
Test Case						
Test Case ID		Date Created		Created by		
TCT_SS				Asma Abdul salam		
Test Case: Aidro Splash Screen						
Test Script Steps/Results						
Step No.	Description of the Test Case	Expected output	Genuine Outcome	Pass or fail?	Date	By
1	Launch the Aidro app.	Splash Screen appear along with the application logo when starting the app	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	11/12/24	Aparna Agarwal

Report Now: A Mobile Solution for Reporting Road and Electricity Issues

Test Case

Test Case ID	Date Created	Created by
TCT_SS		Asma Abdul salam

Test Case: New Account Screen

Test Script Steps/Results

Step No.	Description of the Test Case	Expected output	Genuine Outcome	Pass or fail?	Date	By
1	User press register button when Full Name, Mobile, E-mail and Password are blank	Show error message to notify user to fill-in all required fields	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
2	User write down different password in repeat password field than the one in password input field	Show error message to notify user that password are not matched	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
3	User press register button but the email is already exist	Show error message to notify user that there is an account linked to this email and suggest to login instead.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
4	User press register button when email or password are blank	Show error message to notify user to fill-in all required fields	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal

Report Now: A Mobile Solution for Reporting Road and Electricity Issues**Test Case**

Test Case ID

Date Created

Created by

TCT_SS

Asma Abdul salam

Test Case: Login Screen**Test Script Steps/Results**

Step No.	Description of the Test Case	Expected output	Genuine Outcome	Pass or fail?	Date	By
1	User press login button when E-mail or Password are blank	Show error message to notify user to fill-in all required fields	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
2	User click on Register New Account button	Take the user to New account screen	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
3	User enter wrong login credentials	Show error message to notify user that the password or the email are wrong.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
4	User Click on login button all input are correct	Successfully transfer the user to the home screen.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal

Report Now: A Mobile Solution for Reporting Road and Electricity Issues						
Test Case						
Test Case ID		Date Created		Created by		
TCT_SS				Asma Abdul salam		
Test Case: New Report Screen						
Test Script Steps/Results						
Step No.	Description of the Test Case	Expected output	Genuine Outcome	Pass or fail?	Date	By
1	User Click on governornate select box	Display a list of all of Oman Governornates	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
2	User press send button when address input field is empty	Display an error message to notify the user to enter all required fields.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
3	User press capture button	Show Camera to take a picture	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal
4	User captured a photo successfully	Display a message to tell user the photo is attached successfully	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Aparna Agarwal

5	User press get location button	Display a popup to tell user to give the mobile app permission to use GPS location	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
6	User give the app permission to use GPS location	Display a message to tell user that location has been recorded successfully	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
7	User don't give the app permission to use GPS location	Display a message to tell user that location has not been recorded.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
7	User press send button with all fields filled correctly	Go to home Screen	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal

Project: Aidro Mobile Application						
Test Case						
Test Case ID		Date Created		Created by		
TCT_SS				Asma Abdul salam		
Test Case: Admin Screen						
Test Script Steps/Results						
Step No.	Description of the Test Case	Expected output	Genuine Outcome	Pass or fail?	Date	By
1	Admin select one report and press View button	Report details should be shown	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
2	Admin press add feed back button	Popup with input field should be shown	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
3	Admin press save button without enter a feedback	Show error message to notify admin to fill in all required field.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
4	Admin press save button after write a feed back	Successfully close the popup and transfer the admin back to report details screen.	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal

5	Admin press open on Google maps button	Should open Maps app to the report location	☒ As per expected result ☐ Not As per expected result	☒ Pass ☐ Fail	1/1/24	Apama Agarwal
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5.4.4 Test cases analysis

The responsiveness of the application and the database has been tested, as have the system's components and methods of communication, and it has been established that the application functions meet all requirements. As a result, all system functions operate as expected

5.4 Deployment

In the current technical community, developers and software owners deploy both software updates and bug fixes, with testing and installation being the most crucial software deploying activities. Software deployment is defined as making arrangements to guarantee that the software product is delivered to the intended users. It also includes giving users the necessary updates.

5.4.1 User device specification:

The requirements for installing the application on users' phones are not too complicated: in addition to having coverage, the user needs to have a smartphone with a global positioning system (GPS) and an Android or iOS phone. Additionally, the phone's Internet connection is necessary. The user must finish this procedure. In terms of the user's phone's technical specifications, modern processors are able to run the application with ease, and random access memory (RAM) is more than enough to provide the best performance possible for the application.

5.4.2 User experience:

Thanks to the user interface, which has made the application's operation simpler and easier to handle—especially given that it's a notification application—all users can learn how to use it without the need for any extra practice or instruction.

5.4.3 user device specification

user device specification

Processor	Quad core mobile processor and up
ROM	16 GB and up
RAM	2 GB and up
Operation system	Android 5.0 (API 21) or newer

5.5 Critical evaluation

5.5.1 Critical evaluating the system

This project's primary focus was on traffic-related fatalities and accidents. The realistic database from Firebase and the programming logic employed served this purpose clearly and successfully. This made it easier for the appropriate authorities to respond to the issue and deal with it quickly because all the necessary information was filled in, making all the details—including the problem's location and urgency—known. The issue and an image of the issue are provided so that those who are concerned can view these issues. This application is crucial for preserving the safety of street users as well as for safeguarding vehicles, passengers, and pedestrians.

5.5.2 Illustrating the test results:

The login page and signup page work as expected •

The appearance of the user's email on the home page confirms that the user is •
connected

Reporting a problem works well •

- The administrator has access to authentication details.
- Administrators can get the email and password from the admin.
- Enable location-based tracking

The focus of this test was to receive road problems •

- All functions work as intended in the application, including basic functions.

Chapter 6 Conclusion

6.1 Conclusion

The primary objective of this project was to indirectly provide high-quality streets by focusing on reducing the problems that exist on the streets, whether for locals or visitors to the country. This was done by reporting problems on the streets and entering data related to those problems and their specifics so that the appropriate authorities could act quickly to solve them and protect lives while providing high-quality streets, this initiative sought to address a widespread issue that is thought to be extremely harmful and to have negative effects. This project looked for a fix that will greatly lessen this issue and will the application that will be used by the individual who reports the issue will help find the issue. The accounting body will then monitor the website to make it easier for people to access the issue and will fix it to prevent future issues. The pedestrian, the driver, or any other road user is served by this program. This idea may be developed to cover all road issues and service all national streets in all provinces and regions.

6.2 Recommendation

The program can be developed to be comprehensive for all platforms, to include all types of devices, to add more information and special details to include all method-related problems, and to implement user authentication to prevent false reports by confirming the user's identity via phone number or other means. To efficiently detect issues and address them, create a classification system for various problem kinds. To keep the user updated on the status of the issue he reported, turn on the notification system.

6.3 Limitations

This English-language application is appropriate for those who speak it. People who do not speak English will find it difficult to use this application because they do not understand how to use it. Furthermore, in certain locations there is no Internet connection, which prevents users from logging into the programme and reporting issues.

6.4 Impact of the project

6.4.1 Social impact

It strives to address the issues plaguing the streets and enhance their general condition. The infrastructure and roadways that are kept up help to raise the standard of life, lower accident and injury rates, and increase public safety while also assisting the government with the most pressing recurrent issues and long-term infrastructure planning. Give the street repairs enough funding and more effective resources.

6.4.2 Ethical impact

This project has been morally justified, and this application was created in accordance with social norms about privacy and ethics. Additionally, it protects user data and restricts access to only those who can be trusted. Increasing civic engagement has favorable ethical effects. Addressing ethical issues is also crucial to ensuring consistency with ideals and principles.

6.4.3 Legal impact

There are no legal issues with this project. All images and papers are preserved as references, and all definitions are listed together with their primary sources. This project is completely legal and problem-free in every way.

6.4.4 Sustainability aspects

This application includes elements of sustainability through street problem reporting systems that support social and environmental sustainability, play a significant role in encouraging sustainable behaviors, and raise the effectiveness of infrastructure. This is accomplished through clever infrastructure design, since decisions for infrastructure upgrading are informed by information about traffic issues. Infrastructure and its upkeep guarantee resources for long-term, sustainable solutions.

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literature review **WEEKLY PROJECT DIARY**

A):

Student name	Asma abdul salam Alkawahi
Student ID	19f19336
Project title	Mobile application for roads and electricity reporting
Week no	6
Date	12/4/2023

B):

Tasks	Chapter 2 discussion was done
Work done	Start working on chapter 2
Remarks	No remarks

WEEKLY PROJECT DIARY

A):

Student name	Asma Abdul salam Alkawahi
Student ID	19f19336
Project title	Report No: A Mobile Solution for Reporting Road and Electricity Issues
Week no	WEEK 7
Date	19/4/2023

B):

Tasks	Chapter 2 .
Work done	Complete working on chapter 2.
Remarks	There are no remarks.

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WEEKLY PROJECT DIARY

A):

Student name	Asma Abdulsalam Alrawahi
Student ID	19f19336
Project title	Report No: A Mobile Solution for Reporting Road and Electricity Issues
Week no	WEEK 7
Date	19/4/2023

B):

Tasks	Chapter 2 .
Work done	Complete working on chapter 2.
Remarks	There are no remarks.

Page 1 of 2 62 words English (United States) Text Predictions On Accessibility: Good to go

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Project Poster



REPORT NOW: A MOBILE SOLUTION FOR REPORTING ROAD AND ELECTRICITY ISSUES

data analysis - Asma Abdul Salam Alrawahi 19319336

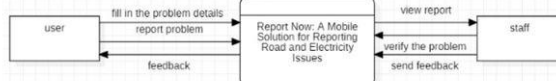
Project Objectives	Rationale of the project
1) urging motorists to report any issue that prevents them from moving forward will increase road safety. 2- Increasing visibility and safety for drivers through awareness of poor lighting conditions and prompt resolution in order to ensure the security of other road users and to be able to identify any potential obstructions. 3- improving the speed of street problem solving. The companion webpage in the program allows the competent authorities to access the issue site promptly. 4- Accident reduction: Through the changes implemented by the responsible authorities and their assistance in resolving issues, this results in a decrease in the number of accidents.	An application was designed and developed that works to find street problems easily through road users, whether they are drivers, passengers, or pedestrians. The mobile application is designed to meet the requirements that can help road users
Methodology	Project client:
the spiral model because it is adaptable, compatible with the project's quality and type, and works on risk analysis and significantly lowers the failure rate of large software projects through repeated risk assessment and verification of the product under development at each stage. Additionally, it lessens the likelihood of mistakes and the time required for re-implementation of the spiral model.	The project targets drivers, pedestrians and passengers on various streets and roads.
Resource:	Risk of project
Computer Mobile phone Database Internet	Knowing the risks is the first stage of avoiding them. Therefore, the estimated risks have been identified as much as possible and appropriate solutions have been developed. .Problems cannot be accessed due to incompleteness of the required data .The problem is that the problem cannot be accessed due to the lack of a network or the wrong location

Limitations of my project:

Constraints exist in all projects, and no project is free of them. Knowing them and trying to develop them is one of the project goals

- The user did not complete the problem information

There is no direct connection in poor coverage areas



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